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Review Article

Review on recent optimization strategies for hybrid renewable energy system with hydrogen technologies: State of the art, trends and future directions

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HIGHLIGHTS

- A text mining-based software called VOSviewer has been adopted to analyze the scientific landscape of the literature body.
- Critically reviewed existing optimization techniques and software tools utilized for the optimization of HRES-H_2.
- Bibliometric analysis has been conducted to investigate the current trends and potential future research topics.
- An insightful discussion has presented enlightening shortcomings and opportunities of clean energy and hydrogen globally.

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ABSTRACT

The world is currently facing a power shortage due to the inadequacy of conventional energy sources and increased energy requirements in almost all sectors of human life. To mitigate this issue, the researchers have taken the considerable interest of researchers over the past decade in enhancing energy efficiency and viability. A hybrid renewable energy system (HRES) can efficiently produce clean energy to meet energy demand. Thus, it is extensively employed to improve power system quality, reliability, and economy, rather than solely relying on non-renewable energy sources. Nevertheless, RE sources' uncertain and intermittent nature, like wind speed and solar radiation, is associated with HRES. This problem can be solved with proper optimization by coupling HRES with energy conversion and storage devices, e.g., electrolyzer, fuel cell, and hydrogen tank, which can admirably balance power generation and energy demand. The literature is rich in employing optimization techniques on HRES with hydrogen technologies (HRES-H_2). However, a gap is found in the overall research progress of optimization approaches, considering HRES coupled with H_2 equipment. Therefore, the current study comprehensively reviews all the optimization approaches applied in this field worldwide. Further, a text mining-based software VOSviewer is used to investigate the scientific landscape of the literature body to figure out the current trends and future scope of HRES-H_2. It has been investigated that the researchers are focusing on: techno-economic optimization of HRES-H_2, developing sophisticated hydrogen infrastructure to reduce the overall cost of hydrogen fuel, introducing AI-based multi-objective optimization techniques to make the HRES-H_2 system more reliable and economically viable, and the impact of renewable and hydrogen

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technologies on the reduction of global warming. Lastly, an insightful of the current review highlighting the present shortcomings and opportunities of clean energy and hydrogen has been discussed, and suggestions are provided.

Contents

Nomenclature

Abbreviations

ABC	Artificial bee colony
ABSO	Advance bee swarm optimization
ACO	Ant colony optimization
ACC	Annual capital cost
BBC	Big bang crunch
BDG	Bio diesel generator
BLMI	Bi-level mixed-integer
COO	Cost of operation
COEG	Cost of energy generation
COES	Cost of energy supply
CSA	Crow search algorithm
DCHSSA	Discrete chaotic harmony search-based simulated annealing
DEA	Differential evolutionary algorithm
DER	Distributed energy resources
DG	Diesel generator
DPPS	Deficit power probability of system load
DSA	Discrete simulated annealing algorithm
FPA	Flower pollination algorithm
GIHRES	Grid independent HRES
GT	Geo thermal
GWO	Grey wolf optimization
HCHSA	Hybrid chaotic/harmony search/simulated annealing algorithm
HEC	Hydrologic engineering center
HRSS	Heat recovery steam system
HSA	Harmony search algorithm
HGWOSCA	Hybrid grey wolf optimizer sine cosine algorithm
HSNPC	Hybrid system net present cost
IACO	Improved ant colony optimization
IAEO	Improved artificial ecosystem optimization
IBA	Improved bee algorithm
ICA	Imperialist competitive algorithm
IHSA	Improved harmony search algorithm
IRR	Internal rate of return
ISCA	Improved sine−cosine algorithm
ISSOA	Improved salp swarm optimization algorithm
JA	Jaya algorithm
LOLIP	Loss of load interruption probability
LAB	Lead acid batteries
LCOE	Levelized cost of energy
LDP	Load deficit probability
LHSP	Loss of hydrogen supply probability
LIP	Load interruption probability
LSCS	Lifespan cost of the hybrid system
MG	Micro Grid
MFFA	Modified farmland fertility optimization algorithm
MILP	Mixed-integer linear programming
MINLP	Mixed integer nonlinear optimization problem
MOGA	Multi-objective genetic algorithm
MOGSA	Multi-objective genetic search Algorithm
MOO	Multi-objective optimization
MOPSO	Multi-objective particle swarm optimization
NRU	Non-renewable usage
NSGA-II	Non-dominated sorting genetic algorithm
PPMS	Predictive power management strategy
RAS	River analysis system
REP	Renewable energy portion
REPG	Relative excess power generation
ROD	Reverse osmosis desalination
RPMS	Reactive power management strategy
SATS	Simulated annealing and tabu search algorithm
SCA	Sine-cosine algorithm
SOC	State of Charge
SPEA-II	Strength Pareto evolutionary algorithm
TCS	Total amount of CO_2 emission
TCR	Total cost rate
TD	Tidal
TLCC	Total life cycle cost
VRF	Vanadium redox flow
WOA	Whale optimization algorithm

Symbols and Subscripts

$T_{A,NOCT}$	Ambient temperature at which NOCT is defined
T_A	Ambient temperature (°C)
V_{SC}	Average voltage of a stack of fuel cell
$T_{C,NOCT}$	Cell temperature at which NOCT is defined
U_L	Coefficient of heat transfer to the surrounding
A_E, B_E	Coefficient of the consumption curve (kW/kg/h)
P_{icon}	Converter input power
P_{ocon}	Converter output power
v_{cut-in}	Cut-in speed
$v_{cut-out}$	Cut-out speed
I_e	Electrolyzer current
$\eta_{h\tan K}$	Efficiency of hydrogen tank
η_C	Efficiency of PV
F	Faraday constant
η_f	Faraday efficiency
c^f	Friction co-efficient
I_{FC}	Fuel cell current
m_{H2}	Hydrogen mass flow (kg/h)
T_{htci}	Hydrogen tank compressor inlet temperature
η_{htc}	Hydrogen tank compressor efficiency
P_{hto}	Hydrogen tank outlet pressure
P_{hti}	Hydrogen tank inlet pressure
S_S	Incident solar radiation at standard test conditions kW/m2
S_S	Incident $_{solar}$ radiation at standard test conditions kW/m2
Y_{pv}	Nominal capacity of PV
m'_{H2}	Nominal hydrogen mass flow (kg/h)
N_C	Number of cells in series

N_{cell}	Number of cells in stack	α_P	Temperature coefficient of power (%/°C)
γ	Polytrophic coefficient	T_C	Temperature of the PV (°C)
P_r	Rated power of wind turbine	Ts	Temperature of the PV under standard test conditions (25 °C)
υ_r	Rated speed		
Q_{H2}	Rate of hydrogen generated by the electrolyzer	$V_{h\ tan\ K}$	Volume of hydrogen tank
f_{pv}	Reduction factor	h_2	Wind turbine height
h_1	Reference height		

Introduction

General perspective

Because of population growth, industrialization, and urbanization, global energy and electricity demand is increasing. The world is facing significant difficulties in providing a safe and affordable energy supply to consumers, as significant traditional energy sources deplete nature by emitting harmful emissions [1,2]. Energy consumption is rising faster than global population growth, and it is projected to increase by nearly 50% by 2050 [3]. The trend of energy use from various traditional and renewable sources is a significant measure of resource usage and environmental effects for sustainable development. More than 80% of the primary supply of energy comes from natural gas, coal, and liquid petroleum, which are now rendered as diminishing sources of energy; also, they are the cause of a large amount of greenhouse gas (GHG) emissions, e.g., CO_2, SO_2, NO, etc. [4,5]. Fig. 1 depicts global CO_2 emissions from various fuel types. Furthermore, fossil-fuel energy sources are a leading cause of the faster rate of climate change. To prevent such a devastating situation, the international energy agency revealed that the rate at which the current renewable-energy infrastructure is growing will lead to a 40% reduction in total carbon emissions by 2050 compared to 2021 [6]. In light of these considerations, the growth, implementation, and deployment of low carbon techniques, especially renewable energy harvesting techniques, have risen to the top of the priority list to meet the global energy needs and reduce CO_2 emissions [7–10].

Demand for environmentally friendly renewable energy sources is continuously elevating. Energy obtained from the sun and wind are the best-known electricity sources, and a combination of these sources is being used to increase energy reliability [11,12]. Some other renewable energy sources, for example, hydro, biomass, and geothermal, are often employed to fulfill the base energy demand [13–15]. The principal problem with the use of RE is the non-constant and infrequent nature due to local climate and geography. Further, the power output is influenced by weather and geographical conditions [16,17]. Thus, an energy storage system is required [18–22]. Electricity is typically stored in batteries, but these possess some issues such as energy leakage, making them not a good choice for long-term operations. Furthermore, since the battery energy density is low, it is not recommended for high storage [23–25]. On the other hand, hydrogen (H_2) is regarded as a promising energy supplier for a sustainable future in the use of renewable energy [26]. However, one major

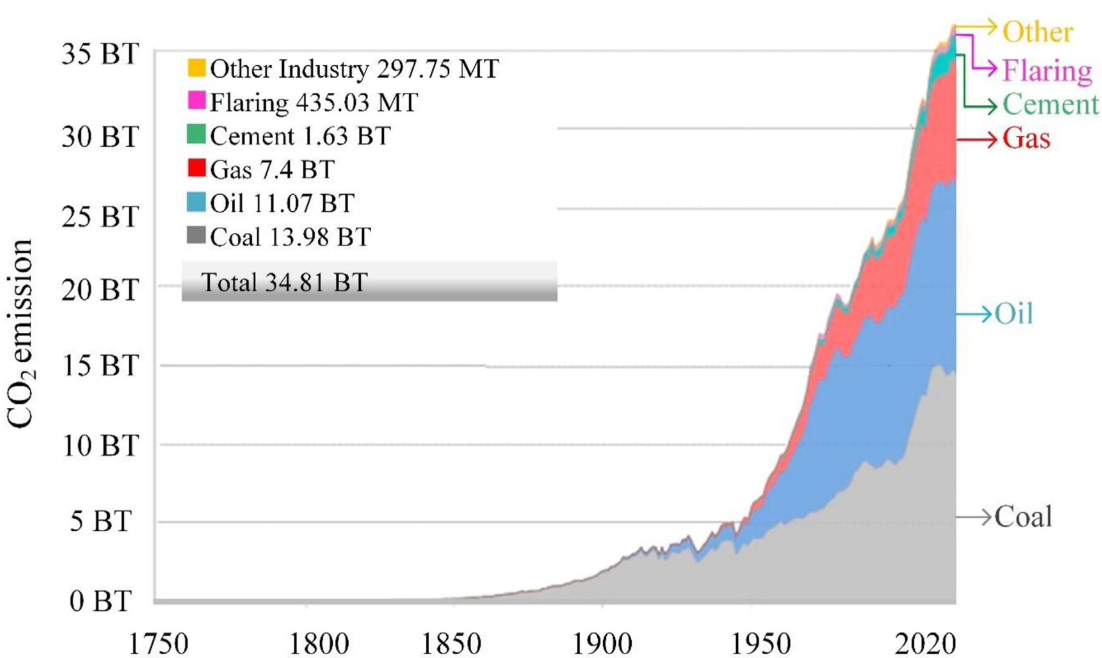

Fig. 1 – Global annual CO_2 emissions from different sources [4].

obstacle to the wider implementation of energy systems based on hydrogen is that hydrogen is not naturally present in appreciable quantities on earth. Hydrogen is obtained from various other sources, such as hydrocarbons or water, through energy consumption [27–31]. Hydrogen production depends substantially on traditional methods, such as steam methane reforming, petroleum refinery, and coal gasification. These approaches contribute approximately 96% of the global supply of hydrogen through processes that generate significant greenhouse gases [32,33]. So, water electrolysis is an alternative environmentally friendly method of producing hydrogen, which allows the use of renewable electricity to produce hydrogen without having harmful emissions [34–36]. As a result, RE reliability, like wind and photovoltaic, can be increased. However, the global production of hydrogen for energy via water electrolysis is currently limited to around 0.03%. This share equals an overall electrolysis capability of 290 MW [32].

Hydrogen as an energy storage vector has a significantly higher energy density than traditional fossil fuels and a considerably low leakage rate than batteries [37,38]. Using an electrolyzer, the surplus energy from RE can be conveniently converted to hydrogen and stored in a hydrogen tank, an energy storage device, without emitting environmentally harmful emissions [39,40]. The H_2 tank inserts energy into the fuel cell (FC) in a race to make up for a power generation deficit. To obtain electric power from hydrogen, FC technology is accepted worldwide as an auspicious, highly efficient, environmentally friendly, and mature power generation technology [41]. Recently, as the FC equipment is evolving, the hydrogen technology has earned more focus from researchers and is thought to have a good future scope. H_2 is now being used to energize the FC capable of long-term energy storage, which clearly shows that the H_2 will supersede the diesel generator systems shortly [42–47]. The FC working principle is opposite to the electrolyzer, i.e., converts hydrogen into electricity [48]. Currently, various types of FC are available such as polymer electrolyte membrane (PEM) FC, solid oxide FC, alkaline FC, etc., among them, PEMFC has gained much attention from researchers due to its light weight and low pressure and temperature working requirement which makes it an ideal candidate for various application such as transportation, backup power, portable power generation, and so on [49–51].

A proton exchange membrane is required for the operation of a PEM fuel cell because it transports hydrogen ions and protons from the anode to the cathode without carrying the electrons that were removed from the hydrogen atoms. PEM technology is constantly improving to make these membranes appropriate for extended operations and even high temperatures [52]. Recently, there are three main families of PEM. The first one is based on polymers of the type of poly(perfluorosulfonic acid) [53]. Nafion has been the most widely used and investigated of these fluorine-containing polymers, and accounts for the benchmark in the fuel cell sector [54]. The thickness of the Nafion membrane has a significant impact on the physical and chemical characteristics of fluorine-containing polymer, as proven in the performance of the iron–chromium redox flow battery (ICRFB) [55]. Thinner membranes were found to be more suitable for cycling operation of ICRFB. Low temperature working capacity (50–90 °C) restricts the use of Nafion membranes, however, efforts are now being made to develop intermediate FC (90–120 °C) and high temperature (140–200 °C) FC. Sulfonated aromatic polymers are the second most common type of conducting material [56]. Sulfonated poly(arylene ether ketone) (SPAEK) and sulfonated poly(ether ether ketone) (SPEEK) are the most commonly investigated sulfonated polymers for fuel cell applications [57,58]. SPEEK membranes are stable enough to work in PEMFCs at intermediate temperatures (120–140 °C). Poly(ether ether ketone) (PEEK) is a semi-crystalline polymer [59]. This nonfluorinated polymer has higher thermal stability and chemical resistance, and its proton conductivity is improved appreciably by sulfonation of the aromatic position, allowing operation at higher temperatures in which electrochemical reaction rates accelerate. The third class of polymers is based on heterocyclic systems, which include the derivates of polybenzimidazole (PBI) [60]. PBI-based polymers have a remarkable resistance to acidic and basic inorganic chemicals, with high glass transition temperatures (425–436°) and strong thermal and mechanical stability. Due to these characteristics, PBI polymers and their derivatives have emerged as prospective candidates for HT-PEMFCs. All of the aforementioned membranes have been utilized in combination with various types of fillers to improve proton conductivity. Since the PEMFC is a complex nonlinear multi-variable system, the modeling of PEMFC generally results in nonlinear parameter estimation problems which usually entail at least more than one. Various authors have proposed optimization algorithms to solve PEMFC model parameter estimation problems [61–63].

Fig. 2 represents the general overview of hybrid renewable energy systems with hydrogen technologies (HRES-H_2). A few concerns are associated with this system, such as high initial cost, the operation cost of energy conversion devices, specification of sites in case of standalone system, and social constraints. All these major factors call for the optimization of HRES-H_2 to provide reliable power to consumers at affordable prices. The optimization of HRES is a complicated task because it requires deep knowledge of energy sources, technological specifications, meteorological data, and load profiles [64–66]. Over the last few years, researchers have attempted to optimize HRES-H_2 using various optimization techniques such as classical optimization techniques, modern optimization techniques, hybrid metaheuristic optimization techniques, and software tools. Classical methods follow a precise methodology to find an optimal solution; however, these approaches have limitations such as rigorous iteration process, inflexibility, passive convergence pace, high calculation time, not responding to dynamic changing, etc. [67]. Therefore, several scientists adopted modern optimization techniques such as particle swarm optimization (PSO), genetic algorithm (GA), harmony search (HS), and others to investigate HRES, primarily combinatorial of solar and wind with hydrogen technologies. Mokhtara et al. [68] adopted PSO for optimization of the photovoltaic-hydrogen system to minimize the COE and CO_2 emissions. Research on power management optimization and sizing of an autonomous WT-PV hybrid energy system coupled with hydrogen ESS was conducted by Baghaee et al. [69] using MOPSO. Siddiqui and Dincer [70] used multi-objective GA to optimize the HRES-H_2 system for producing power, hydrogen,

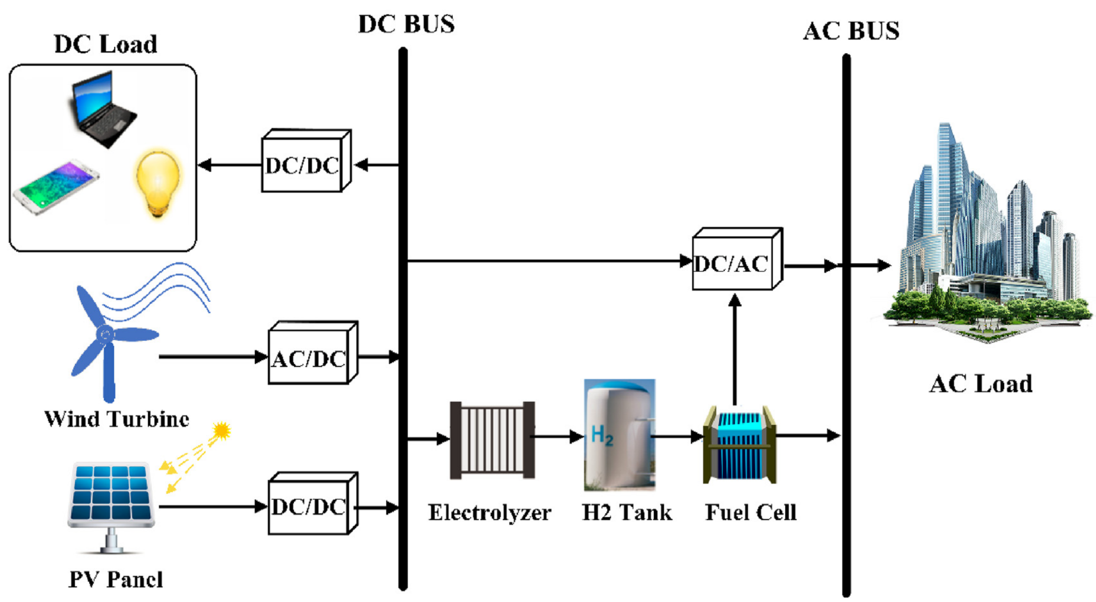

Fig. 2 – Overview of hybrid renewable (PV-WT) energy system with energy conversion and energy storage devices.

and ammonia. He et al. [71] utilized NSGA-II to contrast the technical and economic aspects of HRES with various energy storage systems. Brinda et al. [72] employed the harmony search technique to optimize an autonomous photovoltaic-wind-fuel cell energy system to reduce the capital cost involved in power generation, reduce power losses and CO_2 emission, and improve voltage stability index. Samy et al. [73] analyzed the optimum configuration of various HRES for providing power to rural areas using different optimization techniques such as firefly algorithms, shuffled frolic leaping algorithms, and particle swarm optimization. Fathy et al. [74] have introduced a mine blast algorithm to resolve the PV/WT/FC system's optimal size problem. Since each optimization technique has certain advantages and limitations, authors started to develop hybrid modern optimization techniques, which combine the benefits of individual optimization techniques and avoid their possible limitations. For instance, the simulated annealing algorithm efficiently obtains the optimal global solution, however, it takes more time to locate the best solution in a specific range of possible solutions. Thus, Katsigiannis et al. [75] proposed a hybrid Simulated Annealing-Tabu Search (SA-TS) method for sizing off-grid HRES-H_2. SA is used to determine the suitable zone of a solution, after which tabu search is used to locate the optimum solution. Due to the stochastic and nonlinear properties of chaos variables, chaos search (CS) has been applied in the field of optimization. Chaotic variables can go through all of the states in a particular search region depending on their regularity without iteration, therefore, to improve the search capability of the DHSSA technique, Maleki et al. [76] and Zhang et al. [77] integrated chaotic search with discrete harmony search simulated annealing (DHSSA). Since the searching mode of the firefly algorithm is not systematic, therefore, Samy et al. [78] used firefly algorithm with harmony search HFA/HS optimization technique to obtain the acceptable global solution for hybrid PV/wind/FC system in terms of economy. Jahannoosh et al. [79] applied grey wolf optimizer sine cosine algorithm (HGWOSCA) for optimization of hybrid PV/wind/FC system. The researcher also used simulation tools such as HOMER, RETScreen, TRNSYS, etc., to optimize HRES-H_2. Among them, HOMER has remained in high interest of researchers. HOMER enables investigators to assess various design combinations while taking technical and economic factors into account. Kalinci et al. [80] studied a photovoltaic/wind/fuel cell hybrid system for the generation of electric power and hydrogen by means of HOMER software to maximize the overall system efficiency. Lacko et al. [81] adopted HOMER Pro software to optimally design stand-alone HRES with hydrogen technologies. Singh and Baredar [82] also used HOMER Pro software to conduct techno-economic optimization of the PV/FC/Biomass system.

Contribution of this review

- In the literature, several review articles are available on the optimization of HRES, addressing different issues and aspects of the system [83–90]. It was found that researchers focused on reviewing available optimization techniques of hybrid renewable such as PV/WT coupled with other energy sources (diesel generator) and battery system. However, no review articles are found in recent years delicately and broadly discussing optimization techniques of HRES along with hydrogen technologies. Therefore, to fill this gap, the current review sheds light on the various optimization techniques analyzing economic, reliability, environmental, and social aspects of HRES-H_2 applied by the research community in the past decade.
- Some newly adopted hybrid meta-heuristic optimization techniques in HRES-H_2 in past years have also been reviewed. In addition, a comparison of different optimization techniques, such as modern and hybrid meta-heuristic techniques, has also been presented, which ultimately gives an idea to researchers to carefully select a

suitable technique for their specific design optimization problem of HRES-H_2.

- A range of software has been utilized by researchers to perform simulations for their specific experiments. Each software has its capacity to work in a particular environment of the experimental setup. So, a necessary detail of each software has been included in this review. Moreover, studies conducted on HRES-H_2 with the help of software tools have also been presented.
- Instead of adopting the usual method of searching and analyzing such a huge body of literature which is a time-consuming process, a text mining-based software known as VOSviewer was used to analyze the previous studies. The scientific landscape of literature was developed and analyzed to get a fair knowledge of recent developments and future prospects of HRES-H_2 (Section 5).

The remaining paper is written as:

In section 2, the procedure of selecting and filtering research articles for this review is mentioned. The optimization framework necessary to begin with optimization is introduced in section 3. A detailed literature analysis on the optimization methods, including classical, modern, hybrid meta-heuristic, and software tools, has been discussed in section 4. In section 5, a bibliometric analysis of the literature is presented. Afterward, an insightful discussion is made on the main bulletins of this paper, and future challenges of both the optimization techniques and HRES- H_2 are debated in section 6. Finally, a conclusion is drawn from the current study in the last section.

Mathematical modelling and optimization framework of HRES-H_2

To start with the optimization process, defining a mathematical model is a primary step. Mathematical Modelling is defined as the way of describing the behavior of any system in terms of the abstract model that uses mathematical language [91]. Once the model is developed, the next task is to choose an optimization method or convenient software to solve the research problem. The last task is to compare the optimized results to get the best solution among all possible solutions.

Mathematical modelling of HRES-H_2

PV panel

The photovoltaic power output is estimated by the amount of incident solar irradiations and other environmental factors such as ambient temperature and humidity, etc. PV output power can be calculated by using (1), while the temperature of the PV cell can be evaluated from (2) [92].

$$P_{pv} = Y_{pv} f_{pv} \left(\frac{S_T}{S_S}\right) [1 + \alpha_P (T_C - Ts)] \tag{1}$$

$$T_C = T_A + S_T \left(\frac{T_{C,NOCT} - T_{A,NOCT}}{S_{T,NOCT}}\right) \left(1 - \frac{\eta_C}{\tau\alpha}\right) \tag{2}$$

The variation in the PV cell temperature can be obtained using (3) [93].

$$\tau\alpha S_T = \eta_C S_T + U_L (T_C - T_A) \tag{3}$$

In equation (3), $\tau\alpha$ shows the effective absorption transmittance of the photovoltaic.

Wind turbine

Wind power output is mathematically presented in (4) [9]. The wind turbine begins to generate power as soon as the wind speed surpasses the cut-in value. Further, the wind turbine delivers constant power when the wind speed is greater than the rated speed of the wind turbine. If the wind speed exceeds the cut-out value, the wind turbine generator shuts down to prevent the generator from being damaged.

$$p_{WT}(t) = \begin{cases} 0 & \upsilon(t) \le \upsilon_{cut-in} \text{ or } \upsilon(t) \le \upsilon_{cut-out} \\ P_r \dfrac{\upsilon(t) - \upsilon_{cut-in}}{\upsilon_r - \upsilon_{cut-out}} & \upsilon_{cut-in} < \upsilon(t) < \upsilon_r \\ P_r & \upsilon_r < \upsilon(t) < \upsilon_{cut-out} \end{cases} \tag{4}$$

For more than one wind turbine, the total generated power can be calculated by (5).

$$P_{WT}(t) = N_{wind} \times p_{WT}(t) \tag{5}$$

The wind speed is different at different heights and it can be calculated from (6) at the height where the turbine propellers are rotating.

$$\frac{\upsilon_2(t)}{\upsilon_1(t)} = \left(\frac{h_2}{h_1}\right)^{cf} \tag{6}$$

Electrolyzer

The electrolyzer's rate of hydrogen production is obtained as (7). Moreover, the amount of input electrical energy required by the electrolyzer can be evaluated as (8) [94].

$$Q_{H2} = \eta_f \left(\frac{N_C I_e}{2F}\right) \tag{7}$$

$$I_E = A_E.m_{H2} + B_E.m'_{H2} \tag{8}$$

Hydrogen tank

The power required to compress the hydrogen inside the hydrogen tank is presented in (9). Further, the pressure inside the hydrogen tank is estimated as (10) [95].

$$P_{com} = \left(\frac{\gamma}{\gamma - 1}\right) R \left(\frac{T_{htci}}{\eta_{htc}}\right) \left[\left(\frac{P_{hto}}{P_{hti}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right] Q_{H2} \tag{9}$$

$$P_{\tan K} = \left(\frac{RT_{htci}}{V_{h \tan K}}\right) \eta_{h \tan K} \tag{10}$$

Fuel cell

The output power of the PEMFC stack and efficiency of PEMFC can be evaluated by using (11) and (12) [96].

$$P_{FC} = V_{stack} I_{FC} = V_{SC} N_{cell} I_{FC} \tag{11}$$

$$\eta_{FC} = \frac{P_{FC}}{m_{H2} HHV_{H2}} \tag{12}$$

Converter

The power output of the photovoltaic panel and fuel cell is DC, however, the load is usually AC. Thus, the converter is needed for the conversion of DC power to AC power. Also, the converter is utilized in the power system to smooth the flow of energy. The efficiency of the converter can be determined as (13) [97].

$$\eta_{con} = \frac{P_{ocon}}{P_{icon}} \quad (13)$$

Optimization framework

In this section, the framework of the optimization problem is briefly explained. The flow chart of optimization formulation process is shown in Fig. 3. Each step of the optimization process, from the selection of design variables to finding a solution, is crucial to obtain reliable and optimized results. As the optimization begins, the required conditions of the proposed HRES-H_2 must be satisfied at every succeeding step, otherwise, the preceding steps have to be repeated. In the first step, the design variables are selected, if the developed constraints (second step) are not exactly defining the system then design variables have to be re-selected. The optimization process will proceed to the third after successfully executing the second step. Consequently, in the third step, in case of objective functions are not clearly defined, then the constraints need to be revised until the objective functions are in accordance with the optimization objective. The fourth step is to set up the variable bounds, this step can be redirected to the first and second steps if variable bounds are not satisfying system requirements. The next step is to choose an appropriate optimization algorithm to optimize the HRES-H_2. Any obstacle in this step will redirect the optimization process to all four previous steps one by one. Finally, the optimized results are obtained after successfully implementing the all steps involved in optimization framework. If desired results have not been obtained then the optimization algorithm must be redefined or choose another suitable algorithm. In the next subsections, the optimization framework is further explained.

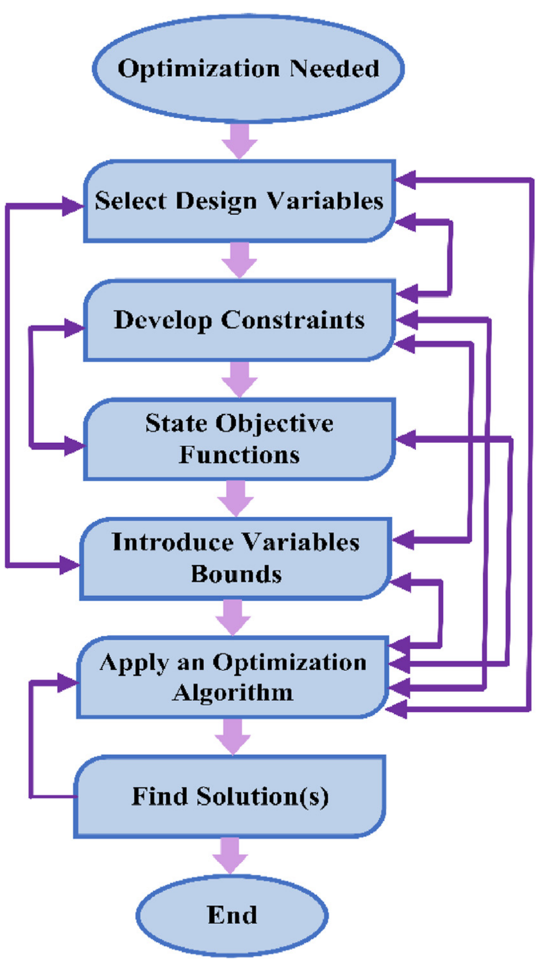

Fig. 3 – Flow chart of optimization formulation.

Variables for design optimization

Design variables are usually defined as control parameters, which are chosen before the optimization technique is employed. They can be single or multiple [98]. The type of design variables defines the optimization problem as either a continuous/discontinuous problem or a hybrid problem, which contains both of the aforementioned problems. In the optimization of HRES-H_2, a range of system combinations exist which includes different types and numbers of equipment such as power production units e.g., photovoltaic and wind, energy storing appliances, e.g., battery and hydrogen tank, and energy conversion devices, e.g., electrolyzer and fuel cell. The size of energy storage devices is described by their capacity and rating, while the size of power production units is determined by their rated power and number. The variables considered by the authors for their optimization studies of HRES-H_2 are available in Tables 3, 4, 6 and 8.

Constraints

In any type of optimization, constraints play an important role in deciding the best solution. There are two types of optimization problems: constrained and unconstrained [158]. Before applying the optimization algorithm to a constrained problem, the constraints which will be implemented to design variables must be defined. Constraints define the search space's boundaries, normally, all design variables are kept between upper and lower limits. The constraints can be either equal or unequal, and they can be simple (maximum and minimum bounds) or complex (dynamic model). Optimization of RES is influenced by component characteristics, and system design, as well as considered objective functions and constraints. Therefore, selecting an appropriate optimization algorithm capable of solving a complicated nonlinear problem is crucial. The constraints considered by the authors for their optimization studies of HRES-H_2 are available in Tables 3, 4, 6 and 8.

Objective function

An objective function must be specified for any optimization process, which typically incorporates economic, technological, and environmental factors [67]. Particularly in view of RES modelling, net present cost (NPC), the annual cost of system (ACS), cost of energy (COE), life cycle cost (LCC), etc. are listed in economic consideration. The reliability objective includes the loss of power supply probability (LPSP), expected energy

Table 1 – A short description of the cost and reliability functions and main formulas.

Cost Indices	Description	Key formulas
NPC [99]	This cost is explained as overall amount of components cost, cost of installation, cost of components replacement, and O&M cost in an entire functional life of the system. All the transactions are changed to the timewhen the investment was first made, taking into account interest and inflation.	$NPC(\$) = \sum_{i=1}^{n} N_i \times \left[(CP_i + RC_i) + \sum_{i=1}^{R} OMC_i \times \frac{(1+i_{inflation})^1}{(1+i_{rate})^1}\right] + C_{inst}$
ACS [100]	It is the amount of the capital cost of the system per year ($A_{cap\text{-}cost}$), annual maintenance cost ($A_{Main\text{-}cost}$), and annual substitution cost ($A_{Rep\text{-}cost}$) of all the components that constitute the whole system.	$ACS(\$/yr) = A_{cap_{cost}} + A_{Main_{cost}} + A_{Rep_{cost}}$
LCOE [101]	This cost is defined as the fraction of annual expenditure to the gross power delivered from the system.	$LCE(\$/kwh) = \frac{ACS}{E_{TAEG}}$
LCC [66,99]	This cost function is the summation of commissioning cost, initial expenditure cost, functioning and overhauling cost, substitution cost and selvage price	$LCC(\$) = \sum_{i=1}^{n} CC_i + RC_i + OMC_i$
Reliability Indices	**Description**	**Key formulas**
LPSP [102]	It is described as the shortfall in electricity production to the energy requirement for a given period of time.	$LPSP = \frac{\sum_{t=1}^{N} LPS(t)}{\sum_{t=1}^{N} LD(t)}$
EENS [103,104]	It is explained as the electricity that's not being provided to satisfy the load demand. It happens due to a rapid rise in demand that exceeds the generation capacity of the system.	$EENS = \sum_{K=1}^{8760} LD$
LA [100]	It is described as the fraction of energy demand satisfied during the functional period of the generating system	$LA = 1 - \frac{T_{LnS}}{T_{Operation}}$
ELF [103]	It is defined as an adequate induced outage duration to the total working hours of the system. When the ELF is less than 0.1, it is considered to be suitable for a standalone system	$ELF = \frac{1}{T} \sum_{t=1}^{H} \frac{E(Q(t))}{LD(t)}$

Table 2 – Merits and drawbacks of classical and modern optimization techniques [82–84].

	Classical optimization methods	Modern optimization methods
Merits	Perform well on a unimodal search landscape Less computational efforts are required Obtain the same result in each iteration Require less function evaluator	Good performance in unimodal as well as multi-modal search landscape. Avoid local solutions Low dependency on the initial solution Obtain the global optimum solution with a higher probability
Drawbacks	Less efficient on a multi-modal search landscape having multiple local optima Highly dependent on the initial solution which led to immature convergence. Less probability to obtain the global optimum solution	More computational efforts are required Slow convergence speed Obtain different solutions in each iteration

not supplied (EENS), equivalent load factor (ELF), etc. [83]. The study of greenhouse gas emissions such as CO_2, SO_2, NO, etc. is usually undertaken with due consideration of environmental aspects. The type of the design obstacle determines whether the design objective is single objective or multi-objective [159]. In most multi-objective optimization problems, usually, a single optimization approach is employed in which two objective terms are added in a summation fashion to obtain an optimized solution. Nonetheless, the different terms in a single objective function, for example, greenhouse gas emission and cost of energy cannot be explicitly merged because of their non-identical nature and meanings. For instance, in this study [160], the system's reliability and the COE are translated into price and weights, which are utilized to declare the different terminology. Likewise, by considering each objective separately, the multi-objective optimization approach provides sets of various Pareto-optimum results in only one simulation [158]. This method allows the selection of components based on the significance of each considered objective [161]. In previous studies, various optimization objective functions have been adopted by scientists in designing RE systems with hydrogen technologies (Tables 3, 4, 6 and 8). In the following section, commonly employed objective functions that are cost and reliability are briefly explained.

Cost function. The objective of sizing renewable energy equipment is to know the definite number of individual equipment which would meet the energy requirement economically considering system design constraints. Various costs were studied in the literature as follows: (reference studies are discussed in Tables 3, 4, 6 and 8).

- The cost of keeping the system components in a good working condition.
- The purchase cost of hybrid system equipment, for example, photovoltaic panel, wind turbine, electrolyzer, hydrogen tank, fuel cell, inverter, among others.
- The expenses of removing device components that are no longer functional.
- The cost of fuel over the system's entire life cycle.

Various distinct methodologies were adopted for the calculation of total system expenditure. It is evaluated by means of net present cost (NPC), annualized cost of the system (ACS), levelized cost of energy (LCOE), and life cycle cost (LCC). A summary of the cost indices and main formulas are provided in Table 1.

Reliability of the system. Reliability objective functions are generally needed to assess the working capability of different engineering equipment. However, keeping in view the reliability of the power system, it is explained as a potential of the power system to satisfy the energy demand adequately and protectively [162]. Adequacy refers to the presence of enough facilities within the power system to meet energy consumption demand or power system limitations, although it excludes dynamic and transient instabilities. These are the facilities needed to generate enough energy as well as the transmission and distribution infrastructure to get the energy to the load locations of potential consumers [163]. Reliability objective functions are adopted to examine the reliability performance of a power system keeping in view a set of specified minimum reliability criteria. In literature, a variety of reliability indices have been considered, including LPSP, EENS, and LOLE are among the most widely adopted reliability indices for evaluating the working capability of HRES-H_2 (reference studies are discussed in Tables 3, 4, 6 and 8). The reliability indices and formulas are provided in Table 1.

Optimization techniques used for hybrid renewable energy system with hydrogen technologies

Optimization methods are usually categorized into four techniques i.e. classical techniques, modern techniques, hybrid meta-heuristic techniques, and software programs/simulation tools. Classical optimization methods employ the iterative method, mixed integer programming, probabilistic approach, graphical approach, and so on [164]. To obtain the optimized solution to a problem, these approaches apply differential calculus [165]. Modern methods employ hybrid meta-heuristic techniques as well as artificial intelligence schemes. Modern methods can efficiently obtain the global optima and acquire fast convergence and good accuracy [36,66,86]. Hybrid meta-heuristic algorithm combines two different optimization algorithms to obtain better results as compared to an

Table 3 – Summary of classical approaches to optimize HRES-H_2.

Ref	Method	Optimized components							Analysis	Objective/Constraint	Variables	Findings	Country
		PV	WT	EL	FC	HT	BS	Other					
[105]	MILP	✓	✓	✓	✓	✓	✓		Economic	Minimize TAC	No of PV, WT, EL, FC, and BT	The cogeneration model was developed to control the bi-directional energy flow of energy repository system. The model can considerably save the overall cost of hydrogen production, particularly under a high load profile.	China
[106]	MILP	✓	✓		✓		✓	MG	Economic	Minimize COE	Capacity of ESS	Time series and feed forward neural networks were adopted to estimate the solar radiation and wind speed. Two analytical models were introduced for grid-connected MG. The problem is designed as MLIP and solved in AMPL. Optimal size of ESS varies for both grid-connected and off-grid MG.	Singapore
[107]	BLMI	✓	✓	✓	✓	✓	✓	GT	Economic and Reliability	Minimize TAC and LCOH	Charging and releasing power of HT and BT, capacity of PV, WT, and EL	To reduce the hydrogen production cost, a model addressing levelized cost of hydrogen was introduced. The model contains a number of continuous and discrete variables to cater issues related to system operation and reliability	China
[108]	MINLP	✓	✓	✓	✓	✓			Economic and Reliability	Minimize NPC and ELF	No of PV and WT, tilt angle of PV storage capacity of HT, output power of RES	Differential evolutionary algorithm was employed to tackle MINLP. Results were compared with PSO which represented the superiority of the adopted algorithm. The overall system cost was reduced, while the reliability of the system was observed to be within an acceptable limit.	Iran

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Table 3 — (continued)

Ref	Method	Optimized components							Analysis	Objective/Constraint	Variables	Findings	Country
		PV	WT	EL	FC	HT	BS	Other					
[109]	MILP	✓	✓	✓		✓		BDG	Economic, Reliability, and Environment	Minimize TAC, COO, BC, power, and hydrogen balance	Area of PV, capacity of WT and BDG	The developed model investigated four different kinds of HRES considering greenhouse gas emission, economy, and grid communication index. Among various HES combinations, PV and BDG systems were examined to have dynamic performance under the same conditions.	Hong Kong
[110]	Iterative method	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize COE and LOLE, power, and H_2 balance	Capacity of HT and BT	PVs were found to be a cost-effective solution as compared to WTs in various regions of Iran. Due to the uneconomical initial cost and low replacement life of FC, it was not employed by the authors.	Iran
[111]	Iterative method	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize NPC	Size of PV, wind, FC, EL, desalination unit, and capacity of HT	Hourly power balance investigation was carried out with iterative simulation. To reduce the system cost, simulation permits to tailoring of the system equipment within a pre-defined quantity.	Tunisia
[112]	Iterative method	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize DPSP, REPG, TNPC, and COE	Power of PV, WT, FC, and EL, volume of HT	The optimal sizing of HRES was done based on techno-economic principle. The improvement of the working methodology showed that the H_2 chain increases the life span of the battery by limiting its excessive use.	Tunisia
[113]	Iterative method	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize COE, COO, and LPSP	Size of PV, WT, BT, No of RES components	PV arrays were observed to be a more economical solution for the desired location than WT. Chemical H_2 production was found to be more efficient than electrolyzed H_2 concerning storage.	Taiwan

[114]	Iterative method	✓	✓	✓	✓	✓	Economic	Minimize TAC and initial cost	No of components	PV power system with different combinations of energy storage devices like battery and FC are studied concerning cost and efficiency of a system and proposed an optimal configuration (PV/BT/FC) of a system.	China
[115]	Iterative method	✓	✓	✓	✓		Economic and Reliability	Minimize COE and TNPC, power balance	Capacity of FC and size of HT	The simulation process of the solar hydrogen energy system was analyzed using MATLAB based on the empirical information of electric power load, solar irradiation, and ambient temperature. The energy distribution, which is the financial factor, levelized the electric power cost by \$0.195/kWh.	Iraq

individual algorithm, but it offers great complexity to code [67]. Apart from the computer software program, HOMER is extensively adopted by researchers for HRES optimization purposes [166]. In few studies, researchers also used RETScreen [156], TRNSYS [157], and HOGA [93]. Fig. 4 represents the categories of optimization methods of HRES-H_2 reviewed in this study. In next section, a review of recently used optimization techniques is presented.

Classical optimization approaches

Classical optimization approaches are deterministic in nature and useful to find a suitable solution for continuous and differential functions by utilizing differential calculus techniques. These approaches employ linear [106] and nonlinear [167] programming, iterative method [168], probabilistic approach [169], analytical approach [170], graphical construction approach [171], and so on. These optimization approaches seem not to be reasonable for problems having multiple local optima and hence not able to converge to global optima [98]. To neglect this issue, the algorithm needs to be rerun several times by randomly choosing the initial condition. Therefore, the result might not be the best global solution. The merits and drawbacks of classical and modern optimization approaches are presented in Table 2.

Mixed integer programming

In mixed-integer programming (MIP) problems, a few decision variables are strained to be integer values at the global optimal solution. The integer variables play a vital role in optimization issues which can be described and determined. Nonetheless, integer variables cause the optimization problem to non-convex which offers more complexity to solve. With the addition of more integer variables, memory and computation significantly increase. In literature, a few studies have been found using mixed integer programming (MILP, MINLP, and BLMI) in the optimization of HRES-H_2. Researchers mainly focused on economic and reliability aspects which are discussed in the following paragraphs.

MILP problems can be solved by utilizing branch and bound, branch and cut, or branch and price algorithms. All these algorithms are helpful to find out the optimal solution with less possible computational efforts. MILP was used for the life cycle optimization of HRES by Zhang et al. [105]. The authors combined branch and cut algorithm with the branch and bound algorithm to solve the MILP problem. Further, there is a variety of optimizer tools available, for example, CPLEX, AMPL, and GAMS available to rectify the mixed integer linear programming problems which makes it somehow simple to employ MILP in HRES-H_2 application. In Ref. [106], authors proposed MILP, solved by AMPL for sizing of energy storage for micro-grids. The performance of the approach was proved by conducting case studies in which the optimal ratings of energy storage systems for island as well as grid-connected MGs were obtained. Pan et al. [107] adopted a CPLEX optimizer to handle the BLMI model for power to H_2 integrated unified energy system. This bi-level model worked in a way that the upper-level model aims at improving the system economy and optimizes the equipment configuration to meet the regional energy demands. On the other hand, the lower-level

Table 4 – Summary of modern optimization methods to optimize HRES-H_2.

Ref	Method	Optimized components							Analysis	Objective/ Constraints	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	other					
[68]	PSO	✓		✓	✓	✓	✓		Economic, Reliability, and Environment	Minimize COE, LPSP, NRU, and LHSP	Size of PV, EL, FC, and HT	A multi-objective sizing issue was resolved as a single optimization problem with the help of e-constraints defining COE as the primary objective function and LHSP, NRU, and LPSP as constraints of the energy system. Two distinct cases were studied having NRU^{MAX} 100% and 1%. The Grid/PV combination was observed to be economically optimal, while on-grid PV/H_2 was found to be a good choice for the future renewable-hydrogen energy system.	Algeria
[69]	MOPSO	✓	✓	✓	✓	✓			Economic and Reliability	Minimize TAC, LOEE, LOLE, and ELF	No of PV and WT, capacity of EL FC, and converter	The algorithm is employed based on an approximate technique for reliability analysis of HRES which led to shortening the simulation duration. Renewable energy data and load profile were considered to be completely deterministic. Components shortage has a significant effect on reliability indices and the economy of the system.	Iran
[71]	NSGA-II, MOPSO, SPEA, SPEA-II	✓	✓	✓	✓	✓			Economic and Reliability	Minimize LCOE and LPSP	Volume of HT, capacity of PV, WT, EL, and FC	Comprehensive metrics based on hyper-volume were introduced to analyze the ability of four multi-objective optimization algorithms. Overall performance of MOPSO and SPEA was found to be relatively better. Thermal energy storage is an economical alternative for different load profiles, resource levels, and energy storage.	China
[116]	PSO, Jaya, CA,	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize NPC, COE, and LPSP	Operating hours of PV, TB, and FC, No of BT	The obtained results of CA and JA were compared with PSO to find the optimal design of the proposed HRES. PSO and JA successfully obtained a suitable solution, however, PSO was found to have a fast convergence speed. CA couldn't reach the optimal solution.	Saudi Arabia

[97]	PSO, GA	✓		✓	✓	✓			Economic, Reliability, and Environment	Minimize COE, and LPSP	No of WT, FC, EL, BS, and HT and capacity of inverter	Operational influence of P-PMS and R-PMS on economy and environment was studied. System working in P-PMS mode performed well as compared to R-PMS concerning cost, renewable integration, and environment. However, P-PMS was examined to be considerably dependent on the effectiveness of the adopted algorithms. P-PMS could significantly reduce the effect of instantaneous response of FC and EL on sizing and working of the energy system	Australia
[117]	PSO	✓	✓	✓	✓	✓		‘	Economic and Environment	Minimize TNPC and CO_2	Power of DER, capacity of EL and HT	A model was introduced to operate the system responsible for managing hydrogen and electricity. Methane utilization was reduced by 10% in the reformer due to the introduction of an electrolyzer in the system. Employment of H_2 storage further minimized the system cost.	Iran
[118]	PSO	✓	✓	✓	✓	✓			Economic and Reliability	Minimize NPC and LPSP	No of PV, size of WT, EL, FC, and HT	The total cost of the HES including 17 PV panels, 4kw WT, 3 kW FC, and EL each was estimated to be 3654.8for 20 years. Results were contrasted with the differential evolution algorithm, the PSO performed quickly and suggested an optimal configuration of the hybrid system that reduced approximately 10% of the total capital cost.	Mexico
[119]	PSO	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize O&M and LPSP	Size of EL and FC, storage capacity of HT, SOC of BT	Fuzzy logic controller based on PSO was used to optimize the HGPS. Minimized variation in batteries SOC, which in turn guaranteed a more prolonged life of batteries, and SOC was raised by 6.18%. The initial expenditure cost can be reduced by 18%. Operation and maintenance cost and LPSP were reduced by 57% and 33% respectively.	Iran

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Table 4 – (*continued*)

Ref	Method	Optimized components							Analysis	Objective/ Constraints	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	other					
[120]	MOPSO	✓	✓	✓	✓	✓			Economic and Reliability	Minimize TAC, LOEE, LOLE, and ELF	Power of WT, PV, FC, tilt angle of PV, storage capacity of HT	The adopted algorithm intended to determine the three-dimensional Pareto optimum set containing 100 possible solutions by means of 2000 repetitions. The range of Pareto optimal sets is sustained with the help of MOPSO. Yearly simulation of the system having 1-h time steps, exact and approximate calculation of reliability functions was carried out. Multi-objective optimization algorithm showed a good performance to search the globally-optimized Pareto-optimal solution in contrast with the conventional PSO.	Iran
[121]	DMOPSO	✓	✓	✓	✓	✓	✓		Economics, Reliability, and Environment	Minimize NPC, LLP, and CO_2	SOC of BT, storage capacity of HT	DMOPSO was utilized to evaluate a Pareto front (PF) to assist policymakers in choosing a suitable combination of HRES. It was shown that a minimum 67% of Pareto solutions achieved from DMOPSO are better than the solutions given by other existing algorithms. Sensitivity analysis revealed that if the price of photo-voltaic panel and wind turbine is reduced to half, NPC will be reduced by 18.8 and 3.7%.	Spain
[70]	MOGA	✓	✓	✓	✓	✓			Economic and Reliability	Minimize TCR and improve exergy efficiency	Solar intensity, area of PV, swept area of WT blades, wind speed	Tri-generation system producing ammonia, hydrogen, and electricity was studied using MOO technique. Depending on the magnitude of weather data, the exergy efficiency was observed to vary between 10.9% and 38.2%.	Canada

[93]	GA	✓		✓	✓	✓	✓		Economic and Reliability	Minimize TNPC, improve system performance	SOC of BT, pressure of HT	Three distinct energy management schemes and sizing algorithms were used. Each configuration of HES obtained from the aforementioned schemes and algorithms is also investigated using TRNSYS software to confirm its accuracy.	Iran
[122]	FPA, GA, SA, CA, GWO, ICA, ISA, OIA	✓	✓	✓	✓	✓			Economic and Reliability	Minimize COO, COES, and COH_2	SOC of HT	The system ensured to satisfy household load in which thermal demand of the houses was fulfilled using wasted heat of the FC and electric heater. Among various other algorithms considered in this study, ISA delivered the most promising results. Incorrect setting of initial SOC would lead to increase the economy of the supply energy cost as well as decrease the reliability of hybrid energy system.	Iran
[123]	FPA	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize TNPC, LOEE, and LOLE	No of PV, WT, EL HT, FC, and inverter	Power produced by renewable sources and hydrogen energy storage is inversely proportional to the reliability constraints of the system. PV panels were examined to be more economical than wind turbines in selected regions of Iran. Further, wind turbines can be utilized as a backup energy source to meet the load requirement at peak load.	Iran
[124]	FPA	✓		✓	✓	✓			Economic and Reliability	Minimize TAC, TNPC, LCOE, and LPSP	No of PV, WT, BT, and HT	The adopted approach assisted the energy system to provide power in rural and urban locations of Egypt. The optimized system cost and equipment were calculated. Optimization reduced the surplus energy of the system.	Egypt
[125]	ABC, PSO	✓	✓	✓	✓	✓	✓	Hydro	Economic and Reliability	Minimize COE, LOLE, LOEE, and ELF	Capacity of WT, PV, BT, HT, and Hydro	The study was conducted to reduce the overall expenditure and improve the reliability of off-grid HRES-H_2. To confirm the accuracy, the results achieved from ABC were also contrasted with the results of PSO with the help of HOMER software.	Iran

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Table 4 – (*continued*)

Ref	Method	Optimized components							Analysis	Objective/ Constraints	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	other					
[126]	ABC	✓	✓	✓	✓	✓			Economic and Reliability	Minimize TAC and LPSP	Storage capacity of HT, No of HT, convertor/invertor	The algorithm was employed to analyze the case study. When LPSPmax is equal to 0%, 0.3%, and 1% respectively, the PV/WT/FC combination was found to be the most economical system, while at LPSPmax = 2%, the WT/FC was found to be the most economical system.	Iran
[127]	IBA	✓	✓	✓	✓	✓	✓	ROD	Economic and Reliability	Minimize TLCC, and LPSP	Swept area of WT blades, No of HT and BT, SOC of BT	The method was employed to optimally design six different types of off-grid HRES-H_2 along with a reverse osmosis desalination plant to meet the energy demand and for improving fresh water opportunities. GIHRES based battery energy storage was found to be more economical than hydrogen energy storage.	Iran
[128]	SA, TS, HS, PSO	✓	✓	✓	✓	✓	✓		Economic	Minimize ACS	SOC of BT, No of PV, swept area of WT, No of HT	PSO was found to be more robust and results obtained from PSO were more profound as compared to other algorithms applied. Hybrid energy systems having battery storage system are cost-effective option for generating electricity than hybrid energy systems having H_2 tank storage systems.	Iran
[129]	IHS	✓		✓	✓	✓			Economic, Reliability, Environment, and Social	Minimize TLCC, and LPSP	No of HT, area of PV, SOC of HT	Optimal size and location for commissioning H_2 based off-grid PV system in remote areas were studied. Improved HS-GIS produced overall good results e.g., computational time, running cost, convergence pace, etc. as compared to simple HS-GIS. With the increase of LPSP, TLCC of the system was reduced.	Iran

[72]	HS	✓	✓	✓	✓	✓	DG	Economic, Reliability, and Environment	Minimize COE, power loss, and CO_2	Capacity of PV, WT and FC, Bus voltage limit	Algorithm was applied to enhance the result of HRES-H_2 with respect to better VSI, low cost, minimum power loss, and minimized emission in comparison with the existing algorithm for a standard IEEE33 bus system.	India
[130]	ICA	✓	✓	✓	✓	✓		Economic and Reliability	Minimize NPC, LOEE, LOLE, and ELF	PV angle, Storage capacity of HT	Self-governed and grid-connected green energy system were optimally modeled. Self-governed system was found to be less expensive than a grid-connected system but the former one is not environmentally friendly.	Iran
[131]	WOA, PSO	✓	✓	✓	✓	✓	TD	Economic and Reliability	Minimize HSNPC, COE, and LDP	Solar radiation, water flow, and wind speed	Seven distinct combinations of hybrid systems models were employed at three distinct locations in Iran to obtain the optimum configuration considering the low cost and more reliability. Results represented the superiority of WOA in contrast to PSO to find the optimal combination.	Iran
[132]	WOA	✓		✓	✓	✓	BM	Economic and Reliability	Minimize TNPC and LPSP	Area of PV panels, No of bio-waste units, EL, HT, and inverters	WOA outperformed PSO in this design optimization problem considering economy, reliability, and overall efficiency. The number of individual components is proportional to TNPC and inverse to the system reliability.	Iran
[74]	MBA, ABC, CS, PSO	✓	✓	✓	✓	✓		Economic and Reliability	Minimize TAC and LPSP	No of WT, PV, EL, HT, FC, and inverters	Size optimization of hybrid energy system was done using actual modulated data of solar irradiation, surrounding temperature, and wind velocity. Proposed algorithm was compared with PSO, ABC, and CS. MBA will cut down the yearly overall cost of the presented system by approximately 24.8%, while PSO and CS will save costs by around 8.956% and 11.5576%.	Egypt

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Table 4 – (*continued*)

Ref	Method	Optimized components							Analysis	Objective/ Constraints	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	other					
[133]	IAEO	✓	✓	✓	✓	✓			Economic and Reliability	Minimize COE, and LPSP	Power of dummy load and fuel cell, and volume of HT	To increase system reliability and decrease system cost, the model based on MOO was adopted. The model was evaluated on six distinct system configurations including off-grid as well as on-grid HRES. IAEO exhibited optimal performance when compared with AEO PSO, GWO, and SSA. FC contributed the highest overall cost for off-grid HRES, while PV was found to be adding maximum ACC for grid connected HRES.	Egypt
[134]	ISSOA	✓	✓	✓	✓	✓	✓		Economic and Reliability	Minimize COEG and DPPS	Capacity of HT and BS	PV/WT with H_2 was found to have higher COEG than PV/WT having a battery storage system, while the latter shows higher system reliability than the former one. To design a hybrid system, ISSOA showed better performance as compared to SSOA and PSO.	Iran
[135]	JA	✓	✓	✓	✓	✓			Economic and Reliability	Minimize TAC and LPSP	Capacity of HT, No of PV and WT	The optimized solution obtained for both low and high user's load profiles exhibited that for LSP^{MAX} at 0% and 2%, the PV/FC was observed to be a less expensive system in contrast to PV/WT/FC and WT/FC system. Among GA, PSO, Jaya, and backtracking search algorithms, Jaya algorithm was found to be faster.	Pakistan
[136]	CSAadaptive-AP	✓		✓	✓	✓		DG	Economic and Reliability	Minimize TNPC, REP, and LPSP	No of PV, EL, HT FC, and DG	In this method, awareness probability is modified in an adaptive manner. TNPC was found to be increased by 6.4%, when LPSP decreased from 0.01 to 0. In addition, TNPC of the system was significantly affected by the renewable energy portion.	Iran

[137]	DSA	✓	✓		✓	✓	✓		DG	Economic	Minimize TAC	No of PV, WT, and HT, capacity of HT	The adopted approach assisted HRES with DG to provide electricity to a remote area of Iran. The optimized system cost and equipment were calculated. Hybrid photovoltaic/diesel/FC system was observed to be a more economical system considering the fuel cost.	Iran
[138]	MFFA	✓	✓		✓	✓	✓			Economic and Reliability	Minimize LCOE and LPSP	No of PV and WT, capacity of HT and FC	An advanced approach is introduced to decrease the optimization time span. The approach is based on the reliability of the HRES, little variations in the added energy to the grid, and excessive adoption of both wind and solar main attributes. MFFA algorithm reached the minimum COE within 8 iterations, which takes an implementation time of 3664.6 s, while FFA reached the optimal solution in 11 iterations, which takes a time of 6792.7 s.	Egypt
[139]	IACO	✓	✓		✓	✓	✓			Economic and Reliability	Minimize NPC and LPSP	SOC of BT, No of BT, PV and WT, storage capacity of HT and BT	The method is based on sorting, which can assist to minimize the searching time span. It was found that increasing the system reliability by decreasing the system capital cost is ambiguous. It was suggested that more aspects are supposed to be considered for the investigation of model to get the better results. Further, in IACO, the total count of ants chosen as m should be tried many times, and it does not have a scientific basis. A suitable m is crucial to conduct this investigation.	China
[140]	ACO	✓	✓		✓	✓	✓	✓		Economic and Reliability	Minimize TAC and LPSP	No of WT, PV, BT, and HT	The comparative analysis was carried out among the proposed hybrid system with hybrid system having batteries and hybrid system having fuel cells. Because of battery leakage and low efficacy of fuel cell, under some constraints, the presented hybrid system was found to be more economical than others.	China

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Table 4 – (continued)

Ref	Method	Optimized components							Analysis	Objective/ Constraints	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	other					
[141]	ISCA	✓	✓	✓	✓	✓			Economic and Reliability	Minimize CHSLS and LOLIP	No of PV, WT, EL, FC, HT, and invertor, capacity of HT	To find an optimal combination of HRES, ISCA was employed based on a non-linearly decreasing inertia weight system (NDIWS). The comparison of ISCA with SCA and PSO was made considering several combinations and LOLIP varying conditions. ISCA was observed to perform well.	Iran
[142]	AWSA	✓		✓	✓	✓			Economic and Reliability	Minimize TAC, LCOE, and LPSP	No of PV, EL, HT, and FC	Results obtained from the proposed method showed fast convergence and robustness as compared to other optimization methods.	China
[108]	DEA	✓	✓	✓	✓	✓			Economic and Reliability	Minimize COE and LOLE	No of PV and WT, tilt angle of PV, storage capacity of HT	An approximate reliability model was used for reliability assessment. Numerical results depicted that total system cost was reduced as well as system reliability lies within an acceptable limit. The adopted algorithm showed a satisfactory performance when compared with PSO.	Iran

Table 5 – Summary of the advantages and limitations of modern optimization techniques studied in this review.

Ref	Algorithm	Motivation	Superiority	Limitation
[97,116–119]	Particle swarm optimization	Imitates the motion attitude of bird and fish	Searching is fast; simple in calculation in contrast to other optimization algorithms	Not able to perform well while solving problems having no co-ordinate system; potentially gives the partial optimized results
[93]	Genetic Algorithm	Imitates the behavior of natural transformation, e.g., inheritance, mutation, selection, and crossover	Able to find the multiple solutions of a single problem; can be found in MATLAB toolbox; conveniently transfer to other simulations	Cannot quickly converge as that of other meta-heuristic algorithms; optimization time span is not fixed in each run
[123,124]	Flower Pollination Algorithm	Mimics the pollinating behavior of flowers	Can perform both local and global optimization	A higher value of probability degrades the exploitation which leads to diverge the solution from the original one. Lower value of probability reduces the exploitation capabilities.
[125,126]	Bee-inspired algorithms	Based on the smart rummaging attitude of honey bees	Possess quality of both local and global search; can be applied to various optimization problems; is simple to implement; capable of being hybridized with other existing algorithms	Arbitrary initialization; it has various parameters
[72,129]	Harmony search	Based on extemporizing action of jazz musicians	It can favor stable as well as unstable functions; can able to consider discrete and continuous variables; no need for primary value calibration for the variables; no divergence; possesses the quality of both local and global search	Complex solving process
[122,130]	Imperialist competition algorithm	Based on a socio-political planning	Good convergence veracity, suitable for optimization of non-linear hybrid electricity production system problems having multiple dimensions	Complex solving process
[74]	Mine blast algorithm	Motivated by landmine explosions	Able to solve both discrete and continuous optimization problems	Local search ability is a potential weakness; it is a memory less optimizer.
[131,132]	Whale optimization algorithm	Imitates the snaring nature of the whale	High exploration ability because of the position upgrading behavior of whales	Usually, experiences immature convergence which led it to stuck in local optima
[136]	Crow search algorithm	Inspired by the smart behavior of crows	Needs only two control variables i.e., awareness probability and flight length which offers attributes like simplicity and ease of implementation due to less adjustable parameters	Not effective in high dimensional optimization problems which leads to lower precision and less accuracy in optimization
[122]	Grey wolf optimization	Mimics wolf leadership hierarchy and hunting behavior	Able to solve continuous optimization problems in deterministic environments	Suffers from the lack of the population diversity, imbalance between the exploitation and exploration, and premature convergence
[128]	Simulated annealing	Mimics a procedure where metal gets cool and freezes into a minimal energy crystalline structure	Able to solve nonlinear optimization problems; can handle chaotic and noisy data having various constraints; robust and global method; ability to find a globally optimal solution; does not dependent on any specific characteristic of the optimization model	Contradiction between the trait of the solutions and computational time; requisite work is needed to consider various constraints; tailoring of algorithm characteristics can be rather elegant
[75–79]	Hybrid optimization techniques	Introduced by merging two or more algorithms	More precise results; optimized the problem in less computational time in some scenarios; more competitive as compared to other single optimization algorithm	Offer great complexity; difficult to code

Table 6 – Summary of hybrid metaheuristic optimization approaches to optimize HRES-H_2.

Ref	Method	Optimized components							Analysis	Objective/Constraint	Variables	Findings	Country
		PV	WT	EL	FC	HT	BS	Other					
[75]	SATS	✓	✓		✓	✓	✓	DG	Economic	Minimize COE	Fuel consumption, capacity of PV, WT, FC, and BT	SA offers faster convergence in the vicinity of optimum solutions, while TS is an appropriate option in searching the optimal solution in a provided neighborhood. Thus, SA-TS enhanced the solution in view of working capability and convergence speed.	Greece
[76]	DCHSSA	✓	✓	✓	✓	✓			Economic	Minimize TAC	No of WT, PV, BT, and HT, Capacity of HT	HRES was optimally sized to satisfy the energy demand less expensively. The introduced hybrid algorithm has the ability of not being trapped in local optima.	Iran
[77]	HCHSA	✓	✓		✓	✓	✓		Economic	Minimize TLCC	Surface area of PV, Swept area of WT, No of BT and HT	Comparison of the minimum, maximum, mean, and standard values of six different configurations of HRES represented that the introduced algorithm is more pragmatic in contrast to other algorithms used in this study due to lower index readings. Results demonstrated that the hybrid PV-WT system having an electrochemical storage system provides less expensive and reliable energy than a hybrid system having chemical storage.	China
[78]	HFA/HS	✓		✓	✓	✓		DG	Economic and Reliability	Minimize COE, NPC, and LPSP	No of PV, WT, EL, HT, and FC	HFA/HS exhibited good results in contrast to PSO with respect to total electricity production cost, sizing, and convergence period. When LPSPmax is equal to 2%, the number of photovoltaic panels was reduced by 75%, number of wind turbines was not changed, number of fuel cells raised by 66.6%, number of H_2 tanks and electrolyzer was twice.	Egypt

[79]	HGWOSCA	✓	✓	✓	✓	✓	Economic and Reliability	Minimize LSCS and Maximize LIP	No of PV and WT, capacity of HT, tilt angle of PV, FC power	PV/WT/FC was found to be the optimal configuration considering LSCS and LIPmax to satisfy the load demand. H_2 played a crucial part in minimizing the variation in RE and thus the system attained the best reliability. The efficiency of the fuel cell was observed to be inversely proportional to reserved H_2 and LSCS, while directly proportional to system reliability.	Iran

model minimizes the levelized cost of hydrogen to promote the development of hydrogen. It was concluded that the bi-level model can effectively reduce the hydrogen supply price in the planning stage by adjusting the proportion of renewable energy equipment in the system and taking power from the upper grid station.

Due to the non-linear nature of component properties and the dynamic objective function, the challenge of HRES optimization is intrinsically non-linear. For instance, the output energy from wind sources is always fluctuating which has non-linear relation with wind speed. As a consequence, the optimization problem of HRES is solved using non-linear computation [172]. The scientists have endeavored a range of non-linear programming strategies to achieve this goal. Abedi et al. [108] studied a hybrid photovoltaic/wind/fuel cell energy system. Researchers adopted DEA to solve mixed integer nonlinear optimization problems. The comparative study presented the efficacy of the adopted approach.

Iterative method

One of the easiest methods to obtain the optimal value of a variable is to repeatedly calculate various combinations of variables and find suitable objective function values. In this method, the result of the succeeding iteration's objective function is contrasted with the values of preceding iterations' objective functions. The range of design variable values that produce the preferred objective function is chosen, while the remaining combinations of variables are gradually disregarded. The iterative search algorithms can be more effective in a variety of ways, resulting in a diverse set of numerical approaches.

In [110], the iteration approach is used to optimize hybrid photovoltaic/wind/battery/fuel cell system in four distinct regions. The objective of the study was to reduce TAC and improve system reliability in view of LOLE indices. Correspondingly [111], used an iteration approach to optimize PV/WT/FC HES for a desalination system. The investigation intended to find out the optimized arrangement of system equipment, which in turn leads to minimum cost as well as demand satisfaction. A study on the size optimization of hybrid PV/Wind/H_2/battery was done by Khiareddine et al. [112]. The iterative method was utilized for the analysis purpose. It was highlighted that the H_2 chain increases the vivacity of the battery and reduces the COE. Chen and Wang [113] optimized the hybrid energy system for emission-free building using the Iterative method. Authors studied the safety index, reliability index, and system response under various renewable energy equipment.

Limitations of classical methods

For HRES optimization, linear, non-linear programming, different kinds of commercial software, and iterative method have been used. The transformation of a non-linear model of renewable energy generation to a linear model leaves it prone to calculation errors. Therefore, it is necessary to carefully apply linear programming to the HRES optimization problem. Moreover, the values of the coefficients of variables in the linear equation must be determined. The usage of equation solvers such as GAMS, AMPLE, and CPLEX for HRES optimization is necessary to handle enormous amounts of data. For

Table 7 – Outline of some widely used software programs for optimization of HRES.

Software	Components								Input	Output	Availability
	PV	WT	MH	DG	EL	FC	BS	HT			
HOMER	✓	✓	✓	✓	✓	✓	✓	✓	Load direction, source data element information, e.g., capital, maintenance, and replacement expenses	Enhance unit sizing, energy and net capital cost, and a little portion of renewable energy	No cost www.homerenergy.com
RETScreen	✓	✓	✓		✓	✓	✓	✓	Climate data are taken from NASA, fuel, schedule, initial cost, system components	Electricity export revenue, simple payback, compare the energy performance	Freely available http://www.retscreen.net/
TRNSYS	✓	✓	✓	✓	✓	✓	✓	✓	Meteorological data, input ingrained models	Dynamic simulation of electrical energy sources	Not free of cost
HOGA	✓	✓	✓	✓	✓	✓	✓	✓	Information loading, resources input details, elements, and cost factors	Multi-objective improvement, charge of energy, study for addressing energy	Pro variant is free of cost, while the student version is without price.
HYBRIDS	✓	✓	✓	✓	✓	✓	✓	✓	Solar sizing, variety of wind turbines, number and variety of batteries	Cost, percentage of greenhouse gases	–
HYBRID2	✓	✓	✓	✓	✓	✓	✓	✓	Load demand, resource input, initial financing investment and O&M units' cost, units information	Sizing along with economical optimization, expenditure of energy, diffuse fractions of the various hazardous gases, system payoff time period	No cost http://www.ceere.org/rerl/rerl_hybridpower.html

relatively complicated systems, iterative approaches generally take longer to compute than other existing classical methods. This approach optimizes only a single objective, such as the system's cost function. As a result, simulation modeling for the specified dimension of HRES elements can be used to find other crucial parameters such as the system's reliability. The classical technique used for the optimization purpose is influenced by the complexity of the addressed problem, computational efforts needed and time required for the optimization process. Among the classical methods, MINLP, encoded in an appropriate software, appears to be the most suited methodology for optimizing HRES. Classical optimization techniques have not been extensively utilized for the optimization of off-grid HRES with hydrogen technologies. Table 3 summarizes this section.

Modern optimization approaches

Currently, the modern optimization approaches that are meta-heuristic approaches are extensively employed in HRE-H_2 system for different optimization purposes. These approaches have obvious superiority over classical optimization approaches [173]. These approaches possess the quality to solve different power system operations ranging from small to large scale [36,174]. Various modern optimization approaches have been used by researchers, as shown in Fig. 4. Contrary to classical methods, modern methods solely depend on using arbitrary variables to avoid being trapped in local optima. These methods can also find the global optimal solution to the problem with high accuracy.

In order to ensure uninterruptible and reliable power supply to the consumers with minimum possible electricity charges, researchers often do techno-economic optimization analysis of power system which requires an appropriate optimization technique. In this context, HRES with hydrogen technologies is gaining the attention of power system researchers, therefore, enormous studies have been conducted on the techno-economic optimization of HRES-H_2 using modern optimization approaches. Further, investigators have focused on energy management optimization of HRES with the help of modern optimization approaches to save energy, therefore, reducing the cost of electricity along with improving the environmental sustainability. Since all the major equipment used in HRES-H_2 are expensive, secondly, the reliability of renewable energy is often compromised due to the meteorological conditions, thus, these two issues motivated scientists to conduct sizing optimization of HRES before commissioning the HRES project. In coming subsections, a variety of different modern optimization techniques employed for various optimization purposes of HRES-H_2 have been discussed and summarized in Table 4.

Particle swarm optimization

This optimization algorithm was first developed by Kennedy and Eberhart [175]. The algorithm is inspired by the swarm intelligence observed in birds or fishes. PSO algorithm is generally used where multiple solutions to the specific problem exist. In this algorithm, particles of the swarm proceed in an explicit space to find the global optimum velocity and position in the swarm. In every step, the location and velocity of

Ref	Software	Optimized components							Analysis	Objective/ Constraint	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	Other					
[68]	HOMER	✓		✓	✓	✓	✓		Economic, Reliability, and Environment	Minimize COE, LPSP, NRU, and LHSP	Size of PV, EL, FC, and HT	A multi-aimed size optimization problem was clarified as a single aimed optimization problem by means of e-constraints, where COE was considered as the primary objective function, while both NRU and LPSP were defined as constraints. Two different scenarios were studied with NRU^{MAX} 100% and 1%. Grid/PV was found to be the optimal choice for a studied system in view of cost.	Algeria
[80]	HOMER	✓	✓	✓	✓	✓			Reliability	Maximize system efficiency	No of PV, height of WT	The daily average hydrogen production was 1.49 kg/h having energy and exergy efficiencies of 59.68% and 60.26% respectively. Although the system had a total exergy input rate of 13848.18 kWh/d, the part of this disappeared due to exergy destruction in the PV panel and the wind turbines as 35.48% and 28.92%.	Turkey
[81]	HOMER	✓	✓	✓	✓	✓	✓		Economic	Minimize NPC and LCC	Capacity of PV, WT, EL, HT, and FC	Optimal electricity supply was found to be technically viable with 100% RES and hydrogen technologies. Possibility of alternative uses of current industrial hydrogen technology to balance power supply and demand, and the deviation from the numerical results is only 3%.	Slovenia
[143]	HOMER	✓	✓	✓	✓	✓		DG	Economic and Environment	Minimize NPC, COE, and CO_2	Capacity of EL, BT, and HT, No of PV and WT	Among five hybrid systems for supplying electrical requirements, the most cost effective was the WT/H_2/Battery hybrid system, which has a TNPC of US$63,190 and a COE of US$0.783/kWh.	Iran

Table 8 – Summary of software used to optimize HRES-H_2.

(continued on next page)

Table 8 – (continued)

Ref	Software	Optimized components							Analysis	Objective/ Constraint	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	Other					
[125]	HOMER	✓	✓	✓	✓	✓	✓	Hydro	Economic and Reliability	Minimize COE, NPC, LOLE, and LOEE.	Capacity of WT, PV, BT, HT, and hydro	The study investigated to minimize the overall expenditure of hybrid PV/WT/FC/Hydro power plant. Acquired solution by ABC algorithm was contrasted with the solution of PSO which was achieved from HOMER software.	Iran
[144]	HOMER		✓	✓	✓	✓			Economic	Minimize NPC ad COE	No of WT, capacity of EL, HT, FC, and inverter	HES was designed to provide PtG production to fulfill the residential load economically.	Iran
[145]	HOMER	✓		✓	✓	✓	✓		Economic	Minimize NPC and COE	FC capacity	The hybrid photovoltaic/battery system was observed to be the most suitable option due to cost effectiveness and being environmentally friendly, while the photovoltaic/battery/Fuel Cell power system was found to be another option with high cost as compared to the former one.	China
[146]	HOMER	✓		✓	✓	✓	✓	HKT	Economic	Minimize TNPC and LCOE	No of PV, HKT, FC, HT, EL, and BT	The HEC-RAS software was adopted to find the water speed at various distinct positions of a cross section of the river from bathymetry and flow data to find the optimal location. Further, HOMER was utilized to optimize the system.	Ecuador
[147]	HOMER	✓		✓	✓	✓		HRSS	Economic, Reliability, and Environment	Minimize NPC, COE, COO, and CO_2	PV Power, capacity of HT and FC	The economic assessment was performed using HOMER program. It was claimed that this system is not economically viable at present, yet technically possible. The system is interesting on an environmental level as it emits almost zero emissions.	Spain
[148]	HOMER	✓	✓	✓	✓	✓		DG	Economic	Minimize NPC and COE	Potential of PV, WT, BT, and FC	The photovoltaic/battery system was observed to be less expensive and free from pollution. The system could be an alternative option to diesel power system. The FC based system was not recommended as an ideal system due to more economical expenditures.	Malaysia

[149]	HOMER	✓		✓	✓	✓	✓	DG and HKT	Economic, Reliability, and Environment	Minimize NPC, COE, unmet Load, and CO_2	No of PV, HKT, and FC, SOC of BT, types of BT	Study was conducted to confirm the impact of multiple storage systems on HES. VRX battery system offered minimum NPC and COE. The system comprised of H_2 and FC presented the minimum fluctuations. When utilizing the load tracking control, the fluctuation of the minimum SOC considering net cost was found to be higher than the fluctuation observed when using the load cycle control.	Ecuador
[150]	HOMER	✓		✓	✓	✓			Economic	Minimize COE and NPC	PV Power, capacity of HT and FC	The purpose of techno-economic analysis is to reduce the cost of energy. The HES was modeled to satisfy a 14 kW direct current load.	India
[151]	HOMER	✓	✓	✓	✓	✓	✓	DG	Economic	Minimize total cost of system	No of WT, PV, EL, FC, and HT	The profile of load utilization was investigated and optimally modeled through HOMER program. The solution was contrasted with Big Bang Crunch (BBC) optimization algorithm and GAME theory-based GAMBIT Software. Result exhibited that the system is environmentally friendly having a RE proportion of 99%, which include PV (34%), WT (64%), and FC (1%).	India
[152]	HOMER	✓		✓	✓	✓			Economic and Environment	Minimize COE and NPC	Capacity of PV, EL, HT, and FC	The system achieved 97.33–100% reduction in different components of greenhouse emission, while the cost saving was 88% with a return on investment around 41.3%.	Nigeria
[82]	HOMER Pro	✓	✓	✓	✓	✓		BM	Economic and Reliability	Minimize NPC and COE, maintain power balance	Capacity of PV, BT, EL, HT, and FC	The COE of a biomass gasifier production unit, PV and FC crossover energy system was calculated to be Rs.15.064/kWh and TNPC Rs.51, 89003. The electric power in the system was found to be 36 kWh/year with no power shortfall burden.	India
[153]	HOMER Pro	✓	✓	✓	✓	✓	✓	DG	Economic, Reliability and Environment	Minimize COE, NPC LPSP and CO_2	PV angle, height and speed of WT, lifetime of FC, SOC of BT	Variations in RE resources don't have a significant impact on investment policy. Also, 80% reduction in hazardous gasses was observed.	Egypt

(*continued on next page*)

Table 8 – (continued)

Ref	Software	Optimized components							Analysis	Objective/ Constraint	Variables	Findings	Country
		PV	WT	EL	FC	HT	BT	Other					
[154]	HOMER Pro	✓		✓	✓	✓	✓	DG	Economic and Environment	Minimize LCOE, NPC and CO_2	Capacity of HT, tilt angle of PV, FC efficiency, SOC of BT	Supercapacitor energy storage system (SCESS) was used to enhance the working capability of HES based on LCOE and hazardous gasses emissions. The HES with SCESS possesses the RE proportion (68.1%) and 0.346 $/kWh LCOE. 83.2% gas emissions were controlled which in turn saved 814,428 gallons of fuel.	UAE
[155]	HOMER Pro	✓	✓	✓	✓	✓	✓	DG, HKT, BM	Economics and Environment	Minimize NPC, COE and CO_2	Capacity of WT, PV, HKT, and BM	The pure grid extension was not recommended for small and large villages due to the high values of NPC. With the use of stand-alone systems instead of pure extension, about 10–19% of energy losses and 100 to 200 Ton/year of CO_2 emissions are reduced.	Iran
[156]	RETScreen	✓	✓		✓				Economic	Minimize NPC	No of PV, WT, and FC	Suitable tilt angle of PV was found to be 27° and wind turbine having model N100/2500 manufactured by Nordex was analyzed to be the optimal turbine for Sirjan city. Economic aspects of the system were studied under the guideline and schemes for shareholders.	Iran
[157]	TRNSYS	✓		✓	✓	✓			Reliability	Maximize system efficiency, and power balance	PV Power, capacity of HT and FC	The adopted strategy scheduled power delivery to a load under many conditions through photovoltaic and fuel cells. The hybrid system combination was determined with the help of a systematic method without compromising the power trait to reduce the residual power losses.	China
[93]	HOGA and TRNSYS	✓		✓	✓	✓	✓		Economic and Reliability	Minimize TNPC, improve system performance	SOC of BT, pressure of HT	Three different energy management schemes and three distinct sizing techniques have been utilized as well. Every combination of HES obtained from the aforementioned methods is investigated by using TRNSYS software to confirm its validity. The TNC obtained from HOGA software is less than TNC obtained from the HOMER software.	Iran

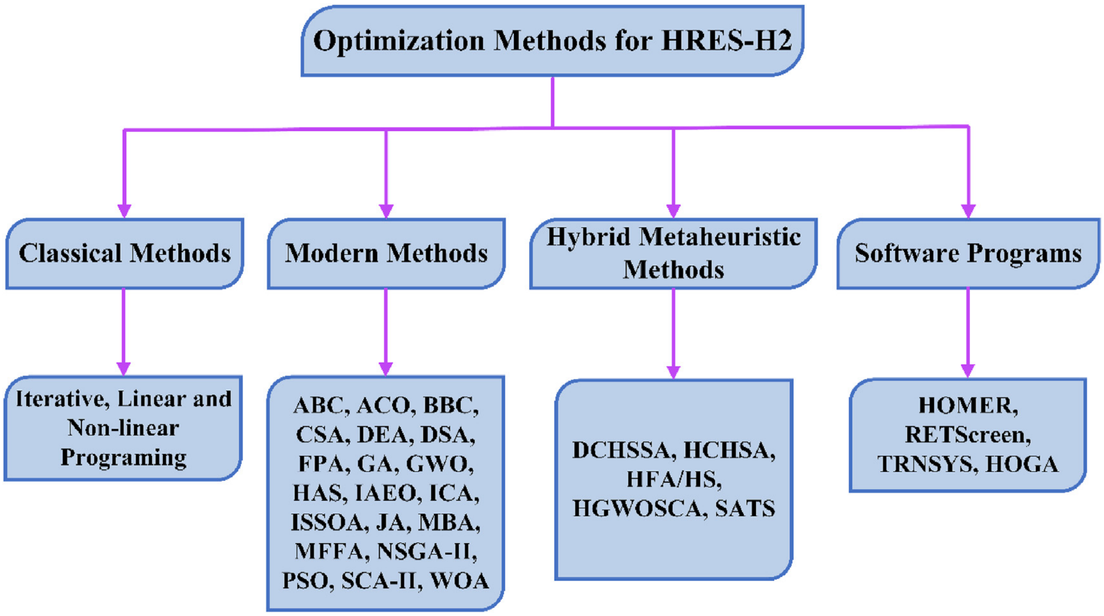

Fig. 4 – Classification of optimization approaches of HRES-H_2 discussed in this review.

every particle of the swarm are calculated to obtain the global optimal number. In literature, PSO has been widely used for techno-economic optimization analysis, energy management optimization, and sizing optimization of HRES with hydrogen technologies keeping in view cost, reliability, and environmental aspects. Tazay et al. [116] investigated technical and economic feasibility of an autonomous HRES. Mokhtara et al. [68] carried out optimization of the photovoltaic-hydrogen system to minimize the COE and CO_2 emissions. Power management optimization for autonomous renewable hydrogen energy systems was studied by Brka et al. [97] using both PSO and GA. Energy management optimization of an HRES established on numerous H_2 production was done by Hassanzadeh et al. [117]. Sanchez et al. [118] and Safari et al. [119] used real solar irradiation, wind speed data, the surrounding temperature, and load demand information to size optimization of an autonomous photovoltaic/wind/fuel cell system.

Most of the real-world problems we solve are multi-purposes in nature, therefore, adopting multi-objective optimization is a reasonable solution to address these issues, which calls for the sound knowledge of optimization. A number of authors also used multi-objective PSO for their studies. He et al. [71] contrasted the technical and economic aspects of HRES with various energy storage systems using four different meta-heuristic algorithms e.g., NSGA-II, MOPSO, SPEA, and SPEA-II. Authors [120] adopted MOPSO to model a hybrid wind-photovoltaic system integrated with H_2 tank to satisfy the load demand for two decades. Research on power management optimization and sizing of an autonomous WT-PV hybrid energy system coupled with hydrogen ESS was conducted by Baghaee et al. [69] using MOPSO.

Researchers faced obstacles in finding solution of complicated problems due to the high dimensional space and nonlinearity of the problem. In this scenario, only optimization approaches are not enough to handle such problems. On the other hand, simulation modeling is capable of adequately handling all system parameters and uncertainties with nearly no hypotheses, but it cannot provide the most appropriate model of the system. As a result, combining simulation model with an optimization approach (simulation-based optimization approach) can be a genuine way to depict the system's complexity, while also delivering the best (or nearly best) results in a fair amount of time. In Ref. [121], a simulation-based dynamic multi-objective PSO algorithm was introduced for the modelling of an autonomous WT-PV-DG-BT-FC system to concurrently reduce the pollution, electricity shortfall, and NPC of the under-study system.

Table 4 presents the summary of articles cited in this section. It is observed that particle swarm optimization and its derivative algorithms are widely famous among scholars when it comes to the optimization of HRE-H_2 systems. The number of research articles on the optimization of HRE-H_2 using PSO and its derivatives has recently increased. Its popularity is because of its capability to find a suitable solution with relatively little effort. Majority of the authors also made a comparison of the results obtained from PSO with other optimization techniques for their specific study to prove the competency of the results.

Genetic algorithm

The concept of genetic algorithm in search optimization was first given by Goldberg [176]. GA adopts the idea of survival of the fittest, which mimics the regeneration mechanism used by biological organisms to deal with their surrounding environment. Selection, crossover, and evolution drivers are used in this algorithm to improve the population quality in coming generations. According to the fitness function criteria, the selection operator selects entities from the present population to be kept for further process. These population entities (variable string) are introduced into the breeding pool, where hybridization occurs, simulating the consolidation of the opposite gender's DNA. These hybrid strings are further arbitrarily mutated. As a result, in a single query, it evaluates the population of several variable values. It leads to a relatively broader search space and a rapid improvement in the population's average fitness. Genetic algorithms possess various merits such as multiple solutions to every single problem, being simple to implement, and can conveniently be shifted to any other simulation, etc. On the other hand, it has a few drawbacks such as convergence towards local optima or any random points despite global optimum solution, optimization time is not constant, etc. Many researchers have utilized genetic algorithm for the optimization of HRES-H_2.

Behzadi et al. [93] employed a genetic algorithm and HOGA software to compare different sizing methods of an autonomous PV/FC/BT system. Authors [97] also used GA along with other algorithms which have been enlightened in section 4.2.1. Rouholamini et al. [122] applied various optimization techniques including GA to discuss optimal energy management of HRES-H_2. After the comparative analysis of different techniques, it was concluded that Interior Search Algorithm (ISA) provides the most suitable optimized results having significant suitability in contrast to various other algorithms. Siddiqui and Dincer [70] used multi-objective GA to optimize the HRES-H_2 system for producing power, hydrogen, and ammonia. In order to reduce the computation efforts, improve the performance and eliminate the need for sharing parameters, and limitations of GA, Deb et al. [177] introduced the variant of GA known as NSGA-II. He et al. [71] utilized NSGA-II to contrast the technical and economic aspects of HRES with various energy storage systems.

GA and its derivative have been used in the literature for techno-economic optimization, size optimization, and energy management optimization. This algorithm obtains the global optimum solution of a problem, having more than one objective optimization problems, with comparatively less computational efforts, which is often experienced in HRES-H_2. The adaptability of this algorithm and efficiency in determining optimal configuration make it a good choice for the HRES. None of the studies found considering environmental and social objective by using a genetic algorithm or its derivatives.

Other modern optimization approaches

Apart from GA and PSO, a range of modern optimization approaches has been observed in the literature for HRES-H_2 design and optimization. To name a few, flower pollination algorithm, artificial bee colony, and harmony search. It is owing to an increase in the volume of scientific progress in the area of evolutionary algorithms, which is developing at a rapid pace with the help of related computer science domains. According to the literature review, scientists are using modern optimization approaches to optimize HRES-H_2, particularly in developing regions such as China and Iran. In coming subparagraphs, modern optimization approaches besides GA and PSO, have been reviewed.

Flower pollination algorithm (FPA) is developed by inspiration of the natural pollination mechanism of flowers, which simulates the performance of pollination. This algorithm is distinguished because of its simple and flexible formulation, as well as its excellent computing efficiency. It has been introduced and implemented in different optimization problems. Intelligent FPA was utilized by Hadidian et al. [123] for sizing and optimum power management of an autonomous HRES-H_2. Results obtained from FPA were compared with teaching-learning based optimization and PSO. Authors recommended FPA for this specific design problem. The authors [122] also used FPA along with other algorithms to discuss optimal energy management of HRES-H_2. Samy et al. [124] chosen FPA to conduct cost optimization of standalone photovoltaic-fuel cell HRES. Researchers also compared FPA with ABC and PSO algorithms, the former presented good performance as compared to the latter two algorithms.

Artificial bee colony algorithm (ABC) is another type of nature-inspired optimization algorithm to solve both unconstrained and constrained problems. This algorithm is motivated by the excellent food finding attitude of honey bees. The artificial bee colony is comprised of three classes of bees: working bees, bystanders, and scouts. The working bees account for half of the colony, while bystanders make up the other half. There is only one working bee for each food source i.e. the number of working bees is proportional to the number of food sources in surroundings of the beehive. When the food source of the working bee is deserted by the other bees, it will become a scout bee. ABC algorithm works in a way that the optimal solution of a problem is indicated by the location of food origin and the grading of the solution corresponds to the quantity of honey. ABC algorithm was employed in hybrid hydro/PV/WT/FC system by Mostofi et al. [125]. Results produced by the ABC algorithm were contrasted with the results achieved by HOMER software. Through satisfied reasoning, the result evaluation proves the accuracy and efficiency of the ABC method in finding the optimum capacity of the combined system. Maleki and Askarzadeh [126] utilized ABC for sizing various combinations of self-governed PV-WT-FC system keeping in view economic aspects.

Bee algorithm is also an evolutionary algorithm developed by the inspiration of the food searching technique of bees. It is firstly proposed by Pham et al. [178]. The biggest difference between the ABC algorithm and BA is that BA uses the elite strategy, more bees are sent to the good quality food source, so high-quality solutions can be achieved and the convergence speed of the algorithm is accelerated. Maleki [127] used an improved bee algorithm (IBA) to design and optimize the (GIHRES) including solar-wind-reverse osmosis desalination systems.

Harmony search (HS) is another type of meta-heuristic algorithm that imitates the extemporizing action of jazz musicians [179]. Individual players in jazz music aim to modulate their pitches so that the entire harmonies are optimal for pleasing sound. They began with a few harmonies and then improvised their way to attain finer harmonies. Rather than harmonies, it can be utilized for the optimization of HRES-H_2. Brinda et al. [72] employed a harmony search technique to optimize an autonomous photovoltaic-wind-fuel cell energy system. The aim was to reduce the capital involved in power generation, reduce power losses and CO_2 emission, and improve the voltage stability index. Results confirmed the performance of the HS algorithm as compared to other algorithms for this specific study. In another study [128], the authors introduced four distinct meta-heuristic algorithms i.e. simulated annealing, tabu search, harmony search, and PSO to find a suitable size of photovoltaic-wind-fuel energy system. Results confirmed the performance of PS considering TAC.

Harmony search (HS) algorithm has fewer mathematical calculations and does not need to define the preliminary values of decision parameters. Further, this algorithm performs stochastic random searches, therefore, differential data is not important. Upon examining all the available vectors, it constructs a new vector. Besides all these advantages, it falls into difficulty, when doing a local search for numerical operations. Improved Harmony search (IHS) algorithm adopts a new technique that enhances the fine-tuning feature and convergence speed of harmony search. IHS algorithm combines the effectiveness of the HS algorithm with the fine-tuning capabilities of mathematical approaches, outperforming both independently [129]. Zhang et al. [129] used IHS and geographic information systems (GIS), for an off-grid PV design taking into account both reliability and economic indices in remote and island regions. It was revealed that the IHS-GIS technique provides better results as compared to the HS-GIS technique in view of the system working cost, robustness, computational time, accuracy, convergence speed, and searching capability.

Imperialist competitive algorithm (ICA) is presented by Atashpaz and Lucas [180]. ICA is motivated by the procedure of the imperialistic race. The algorithm begins having an initial value known as country, which is by default branched into states and territories, forming a kingdom. The imperialist race among many empires makes the root of ICA and in the process of competition time, the successful empires take over the control of their territories. ICA was adopted by Gharavi et al. [130] to model an autonomous photovoltaic-wind-electrolyzer-fuel cell system. It was suggested that the on-grid system is less expensive than the autonomous system with the drawback of harmful gas emissions.

Mine blast algorithm (MBA) was developed in 2012 by Sadollah et al. [181]. This algorithm emulates the process of a series of explosions set in a minefield. In this process, the thrown fragments of shrapnel strike with several other mine bombs around the blast area, causing them to explode. Every shrapnel fragment has specific directions and ranges to hit with other mining bombs, ultimately causing nearby mines to explode, which in turn lead to the identification of severally exploding mine. The fatalities due to the blasting of mine are assumed to be the fitness of the objective function. This process continues till the best fitness value is discovered. Only one article was found in the literature using MBA for sizing autonomous photovoltaic-wind-fuel cell system [74].

Whale optimization algorithm (WOA) was first presented by Mirjalili and Lewis [182]. This algorithm is established on swarm intelligence behavior that imitates the snaring nature of humpback whales. The whale encompasses the prey, once the estimated position of prey is determined. To mimic this, WOA chooses the optimal solution in the closest position to the prey because the prey is situated within the search area and its exact position is not known. After that, the best search agent is used to update the location. Further, the whale makes a narrowing curved path surrounding the prey known as the bubble net action of a whale, and attacks on prey. In WOA, after meeting the convergence requirement and obtaining the desired fitness, the algorithm is ended [182]. Authors [131] did a comparative study for optimization of photovoltaic-wind-tidal-fuel cell at three distinct locations using WOA and PSO. Results depicted that the WOA performed well in contrast to PSO for hybrid system optimization due to less cost and system reliability. Hongyue et al. [132] designed HRES with hydrogen technologies for the purpose to obtain an optimal economy and reliability of the system.

Artificial ecosystem optimization (AEO) was proposed by Zhao et al. [183]. AEO is based on population and developed by the idea of the motion of energy in an ecosystem. AEO method imitates three distinct mechanisms of living beings which are producers, utilizers, and decomposers. Producers (first phase) are an analogy to green plants, which do not consume energy from any organism. The term utilizers (middle phase) is given to animals, which obtain energy from producers or some other utilizers. The decomposer (last phase) relates to the earth, which absorbs plants after their decomposition and waste of consumers. Artificial ecosystem optimization uses the first phase to enhance the balance between utilization and investigation (arbitrarily generating individuals). The middle phase is adopted to enhance the investigation capability of the algorithm (update the solution produced by individuals). The last phase is for the purpose to improve the utilization of algorithm (nominate the best solution among various other solutions). Sultan et al. [133] employed AEO to economically optimize both grid-connected and off-grid PV-WT-FC system. Authors have made a comprehensive statistical measurement to justify the efficacy of the IAEO algorithm. In this study, improved artificial ecosystem optimization (IAEO) is used to improve the performance of AEO in consumption phase.

Salp swarm optimization algorithm (SSOA) is motivated by the swarming nature of salp during exploration and rummaging in the sea. The salps search for the food in the form of a group which is consist of a leader and follower. The salp in front of the chain is the group's leader, while the remaining salps are considered followers. The leader commands the group in search of food, while the followers obey the steps of the leader [184]. Vahid et al. [134] used an improved salp swarm optimization approach (ISSOA) to design a photovoltaic/wind/H_2 system. The comparison analysis shows that ISSOA is superior to SSOA and PSO in achieving the cost of energy generation.

Jaya algorithm (JA) was first presented by Venkata in 2015 [185]. Jaya algorithm needs only the typical control specifications and avoids using any algorithm-based control specifications. Investigators [135] compared the Jaya algorithm with PSO and backtracking search optimization technique to study the optimum sizing of an autonomous PV-WT-FC HES. It was revealed that for PV-WT-FC configuration, JA showed a rapid convergence rate in contrast to GA, back-tracking search, and PSO algorithms. In addition, for PV-FC and WT-FC configurations, convergence rates of algorithms were faster in contrast to photovoltaic-wind-fuel cell energy system due to the smaller number of selected variables.

Crow search algorithm (CSA) is another type of meta-heuristic algorithm and it was first introduced by Askarzadeh in 2016 [186]. The main inspiration of CSA is the smart nature of crows. CSA performs based on the fact that crows save their surplus feast at hidden places and use it whenever required. To improve the searching capability and robustness of the primary CSA, a modified CSA is proposed by Ghaffari et al. [136] for the reliability analysis of HRES with DG. Authors also compared modified CSA with original CSA, GA, and PSO. The result of modified CSA is found to be more reliable.

Sine cosine algorithm (SCA) was first presented by Mirjalili [187]. The SCA establishes various initial arbitrary competitive solutions and influences these solutions to vary outwards or towards the optimal solution with the help of a computational model dependent on sine and cosine geometry functions. Various arbitrary variables were added to SCA to extend the searching and exploitation capacity of optimization. Jahannoush et al. [188] applied improved SCA for optimal modelling and power management optimization of an autonomous HRES. The solution obtained by the adopted ISCA was contrasted with primary SCA and PSO algorithms, which gives favor to ISCA.

SA got its name and motivation from metallurgy's annealing process, which includes heating and cooling a metal to expand the size of its crystals and minimize its shortcoming. SA begins its search with a higher temperature to cover a huge portion of space and ends it with a low temperature (optimal solution). The temperature in SA gradually decreases with the process of iterations. Kirkpatrick et al. [189] firstly proposed the idea of discrete simulated annealing (DSA) algorithm in 1983. DSA is usually adopted when the expected solution is in discrete form. During the search operation, DSA considers both best and worst solutions with some margin of probability. Maleki [137] carried out a study with the help of DSA to model different configuration of photovoltaic-wind-diesel-fuel cell system for various capital costs of fuel prices.

Farmland fertility algorithm (FFA) is based on the farmer's evaluation of the quality of soil in each field zone (optimal solution of a problem). The major cause of differences in soil quality is the presence and addition of specific elements. As a result, farmers utilize a range of products to improve agricultural quality. In other words, when these elements are introduced to the soil, they have the potential to improve or degrade soil quality (available solutions both best and worst). The selection of these elements to improve soil quality is based on the farmers' expertise with each kind of soil and prior improvements in soil quality. FFA was introduced by Shayanfar and Gharehchopogh in 2018 [190]. Further, Ahmed et al. [138] presented a modified version of this algorithm (MFFA) for optimal modelling of a photovoltaic-wind-fuel cell system based on reducing the COE. The solution was contrasted with the solution of the primary FFA and found the good performance of the former one in terms of less computational time.

Dorigo et al. [191] introduced the Ant colony optimization algorithm (ACO) in 1996. The advancement of ACO was impressed by the experience of ant colonies. Ants used to reside in colonies and their living environment is ruled by the aim of colony viability despite being concerned about the life of individuals. In Ref. [139], an improved ACO was selected for the size optimization of PV/WT/battery HES with H_2 as a backup. Optimization of HRES based on battery/H_2 was carried out by Dong et al. [140] with an improved ACO. A simulation experiment was established to represent the obtained results, where comparisons were accomplished to highlight the merits of the proposed system.

Conclusion of modern optimization approaches

A summary of modern optimization methods to analyze hybrid renewable energy with hydrogen technologies is available in Table 4. In literature, a vast variety of studies were carried out on optimum design, sizing optimization, techno-economic optimization, and energy management optimization of HRES-H_2 using a number of meta-heuristic algorithms. PSO possesses a fast convergence speed along with the limitation of immature integration and population diversification. The GA is a suitable option for optimization problems having discrete variables but it has the limitation of being slow in optimization speed in case of a limited search domain. Other than particle swarm optimization and genetic algorithm, FPA, HS, and bee algorithms have been extensively used to figure out the optimization problems of HRES-H_2. These algorithms have the drawback of being locally optimized in view of expanding the magnitude of the problem as well as the investigated system. Due to the aforesaid reason, while determining a solution, inequality is not reasonable between the initial and final values of an algorithm. Thus, in the past few years, keeping in view the limitations of each algorithm, the optimization algorithms have been significantly improved (IFPA, IHS, ISCA, IAEO, IWOA, IGWO, etc.) to rectify their shortcomings. Some merits and demerits of these algorithms are summarized in Table 5.

Hybrid metaheuristic optimization approaches

With the increasing complexity of power system operation, hybrid meta-heuristic algorithms are broadly utilized to solve non-convex optimization problems. Thus, to keep pace with the rapid development of integrated use of renewable energy, more complex and accurate optimization algorithms need to be used, which puts forward higher requirements for hybrid optimization methods. Hence, recently, researchers have made remarkable efforts to develop more sophisticated hybrid meta-heuristic optimization algorithms for the analysis of hybrid renewable energy along with hydrogen technologies. Hybrid meta-heuristic optimization algorithms are primarily a combination of any of the two famous stochastic optimization algorithms. This affiliation between two distinct optimization algorithms gives a worthy characteristic to improve the performance of individual algorithms. In the coming paragraphs, hybrid metaheuristic optimization approaches applied in HRES-H_2 are discussed, which are summarized in Table 6.

The Simulated annealing algorithm (SA) is good at finding the global best solution, but it takes a longer time to find the best solution in a particular zone of a solution. Thus, SA is frequently used in cooperation with local search techniques. The accuracy of the initial solution is critical for the effective employment of the local search technique. The local search technique then iterates from one solution to the next until a specified termination condition is reached. For optimization of HRES-H_2, Katsigiannis et al. [75] developed a hybrid Simulated Annealing-Tabu Search (SA-TS) method. SA is used to locate the optimal region, after which TS is used to find the optimum solution. The presented method enhanced the solution in view of caliber and convergence in contrast to the results given by SA or TS algorithm individually.

Chaos search (CS) has been utilized in the field of optimization due to the stochastic and nonlinear qualities of chaos variables. Chaotic variables have the ability to move through all of the states in a given search region based on their regularity without iteration. Maleki et al. [76] integrated discrete chaotic search with harmony search simulated annealing (DCHSSA) to boost the search capability of the suggested method. The purpose was to obtain the optimal combination of photovoltaic/wind/FC system. Zhang et al. [77] also used this hybrid method to study the HRES including batteries and H_2 tank.

Since the searching mode of the firefly algorithm is not systematic, therefore, it is not able to produce better global results. Moreover, FA requires sufficient time to come up with a better solution, and its performance decreases significantly as the population grows. Hence, it is needed to improve the firefly algorithm to look for a better optimal solution. Samy et al. [78] introduced a hybrid firefly and harmony search (HFA/HS) optimization technique to study the hybrid PV/wind/FC system in terms of economic optimization. The authors contrasted the obtained results with PSO algorithm. The former hybrid algorithm was found to be superior for obtaining the total price of the HRES and convergence time. In order to improve the reliability and pace of the grey wolf optimization algorithm, sine the cosine algorithm based on an exponential decreasing function is hybridized (HGWOSCA), which upgrades the location of each wolf to maintain equilibrium between exploration and exploitation activities as well as improve the convergence of GWO algorithm to find the global optimal solution. Jahannoosh et al. [79] applied grey wolf optimizer sine cosine algorithm (HGWOSCA) for the optimization of a hybrid PV/wind/FC system to satisfy the energy demand of the consumer economically and reliably. The authors suggested HGWOSCA over other optimization techniques for this particular design of HRES-H_2.

Conclusion of hybrid metaheuristic optimization approaches

The innovation of generating the hybrid meta-heuristic algorithm is to specify numerous search methodologies for emphasizing the substitution of the undesired solutions from two or/and three optimization techniques with the desired solutions from the other optimization techniques keeping in view the uncertainty of various parameters. Several efforts have been made to introduce an appropriate combination of more than one optimization algorithm to solve complex optimization problems efficiently. A variety of optimization problems with the objective of minimizing cost and improving reliability have been tackled by researchers using suitable hybrid metaheuristic optimization approaches. However, no study was found taking into consideration environmental and social impacts. It is essential to keep the relative benefits of each algorithm when integrating these algorithms. Hybrid optimization algorithms provide some benefits as observed in studies included in this work: 1) It has better performance in searching for global optimal solutions to a broader range; 2) It is a pragmatic and can eventually find a better solution for various complex research questions; 3) It can effectively solve the problem which is critical to find an appropriate mathematical model; 4) It can avoid immature convergence and unreliable solution of a problem. Thus, the hybrid optimization algorithms would reveal the researchers with competent tools to have remarkable supervision and working of the renewable energy system. Table 6 provides the summary of this section.

Software tools

The suitable combination as well as the optimal working of HRES are dependent on each other and make the system more complex to model, size and investigate. The objective of the optimization of HRES-H_2 is to take maximum advantage of RE sources and hydrogen technologies with minimum cost, which shows that this system can optimally perform well considering cost and reliability. Optimization techniques can assist to assure the optimal results with the smart utilization of equipment. Studying HRES with hydrogen technologies in depth detail is obviously a complex task, due to multiple power production, power conversion, and hydrogen storage systems, which calls for sophisticated optimization methods. Therefore, various software tools such as HOMER, RETScreen, TRNSYS, etc., have been introduced for the various optimization purposes of the HRES. Table 7 shows the detail of the software tools used to analyze the HRES-H_2 system. Sinha et al. [192] contributed the work to review various software tools for the optimization of hybrid system. Literature shows that researchers preferred different optimization software tools according to their specific requirements of the HRES-H_2 system. The summary of their studies is presented in Table 8.

HOMER software

Among many software tools, Hybrid Optimization Model for Electric Renewables (HOMER) is one of the most prominent tools for sizing a hybrid energy system. HOMER enables an electrical system's physical behavior to be determined by its life cycle cost, which is the summation of system components installation and maintenance costs over the lifespan of the system. It is able to perform simulation for one-year data. HOMER allows the researchers to evaluate a large number of different design combinations considering technical and economic advantages. Thus, it is used by several authors to conduct optimization of HRES-H_2 using this software tool.

HOMER can be used to quickly perform feasibility, optimization, and sensitivity analyses in a variety of system combinations. HOMER simulates various system configurations using inputs such as multiple technological possibilities, unit costs, available resources, supplier data, and so on and returns results in the form of a list of possible configurations ranked by NPC. A number of authors preferred HOMER software to optimize HRES-H_2 and conducted feasibility and sensitivity analyses for their particular optimization problems. Fazelpour et al. [143] investigated the feasibility of commissioning a wind/photovoltaic hybrid plant aiming to produce H_2 gas. Feasibility analysis of HRES with hydrogen technologies for domestic load was investigated by utilizing HOMER [81]. Das et al. [148] also carried out feasibility and sensitivity optimization of hybrid PV/battery/FC system using HOMER software. Arevalo et al. [149] conducted a sensitivity analysis to observe the effect of numerous storage devices on HRES with the help of HOMER software. Kalinci et al. [80] studied a photovoltaic/wind/fuel cell hybrid system for the generation of electric power and H_2 by means of HOMER software to maximize the overall system efficiency. Performance study of integrated PV-

FC and diesel based power system having super-capacitor ESS was carried out by Ref. [154] using HOMER software. Pal et al. [150] conducted a viability analysis of DC PV/H_2/FC system using HOMER software. To supply fresh water and electricity simultaneously at five distinct weather zones, authors [155] investigated the selection making strategy between RE configurations and grid extension using HOMER software. HOMER and GAMBIT software were used by Khare et al. [151] to optimize HRES-H_2 for the replacement of traditional electricity generating resources.

RETScreen

RETScreen is a feasibility assessment tool that can be used anywhere around the globe to evaluate the economic viability, energy generation and its efficient utilization, greenhouse gas reduction, and risk assessment of various RE technologies. Off-grid solar panel systems are also covered by the RETScreen photovoltaic module, which includes both standalone and hybrid systems. It includes a worldwide meteorological data database (monthly solar irradiance and temperature records for the year), energy resource data (such as wind mapping), hydrological data, and product information such as solar panel specifications and wind turbine curves. RETScreen can examine data of up to 50 years in monthly or annual time intervals [193]. Mostafa et al. [156] used RETScreen to study PV/Wind/PEMFC hybrid energy system. Authors reduced the IRR of the system in order to make it economically viable.

TRNSYS

TRNSYS was originally designed to simulate thermal systems, but over nearly 35 years, it has evolved and expanded its capabilities. It has evolved into an integrated simulator that includes PV, thermal solar, FC, HVAC, CHP, and other components. It is a graphical based program that can be used to study the transient response of a particular system. It has two parts that are a kernel and a library. The kernel analyses the input data set and resolves the system using different methods and find the convergence, whereas the library contains various components to simulate the behavior of a system that can be updated by the users [194]. TRNSYS performs simulations with remarkable accuracy, including graphics and other aspects. Mubaarak et al. [157] used TRNSYS software for the power management scheme and optimal combination design of PV-H_2-FC system to satisfy the energy demand of consumers efficiently. Behzadi et al. [93] employed TRNSYS, genetic algorithm, and HOGA software to compare different sizing methods of hybrid PV/FC/BT system.

Limitations of software tools

This section introduced the HOMER, RETScreen, and TRNSYS software tools that were employed in HRES-H_2 investigations. The most frequently used tool was found to be HOMER having highest number of RES configurations and dose optimization and sensitivity analysis, making it simpler and quicker to analyze the vast variety of different HRES. In spite of multiple advantages of HOMER software in the area of optimization, it possesses some limitations. It permits just one objective function for decreasing NPC and is not able to formulate more than one objective function simultaneously. Further, HES cannot be prioritized based on LCOE. The drawbacks of RETScreen include limits for energy modelling and the database for building envelope parameters. It does not deal with complex technical problems. The temperature affects the working of PV, which is not considered in RETScreen. For new researchers, the interface of software demands time to work on it efficiently. Simulating a large project in TRNSYS, the task becomes extremely difficult, and some components operate unusually, preventing them from being employed owing to the unusual connectivity required. In addition, it's tough to pinpoint the root of the error. The improvement in the models and increased user customization in these software will improve future research and simulation of HRES-H_2, which in turn reduce the economy of the system through better planning. In addition, more research to demonstrate the potential of software tools for HRES-H_2 optimization can be undertaken.

Bibliometric study using text mining software

Since a vast variety of studies are available on the topic of hydrogen based HRES, it is very laborious to search, filter out and analyze the desired research articles. For the sake of ease in this context, a text mining software called VOSviewer has been developed [195]. To begin with the bibliometric analysis, the three main steps of the VOSviewer were followed: (1) publications imported from the search database i.e., Web of Science; (2) based on the imported publication data, the keywords co-occurrence map is created; (3) making different clusters based on co-occurrence map. After taking these three steps, finally, the clusters were analyzed. A huge number of articles can be easily and systematically analyzed using this method.

Articles selection

To conduct bibliometric analysis, publications were collected using one of the most reliable databases, Web of Science. All the information of relevant research articles, for example, research title, abstract, index keywords, publication year, publisher name, etc., were extracted as a tab-delimited text file which is a prerequisite step to further process the literature analysis in VOSviewer. Initially, 954 articles were found which were further reduced by using the built-in filter function of the Web of Science. Articles from pure sciences were omitted after carefully reviewing their irrelevancy to the current topic. Finally, 554 articles remained for the intended analysis published between 1996 to December 2021.

Analysis method

In order to get the overall perspective of research progress, recent trends, and future scope of HRES-H_2, the bibliometric analysis of publications of the last two and a half decades was carried out by two methods that are keywords analysis and overlay visualization analysis of co-occurred keywords. In the first method, VOSviewer takes keywords available within the publication title, the body of abstract, and index keywords which lead to the scientific landscape of the publications representing the overall research progress of HRES-H_2. In the second method, the scientific landscape is constructed based on (1) timeline of the co-occurred keywords from 2016 to until now, and (2) the average citations of the articles containing that keyword. In this way, the current development and future scope of HRES-H_2 were analyzed. Each method is discussed in the following section.

Keywords analysis

To construct a scientific landscape of literature using keywords, all the keywords are obtained from the titles, abstracts, and indexes of the compiled articles. On the first hand, 1484 keywords were observed which were further reduced by a minimum of 10 co-occurrences. Afterward, these keywords were extracted via the built-in automated text-mining option of VOSviewer. Few irrelevant, repeated, and plural words were filtered out by making a thesaurus file of keywords. Based on csv delimited file and thesaurus file, VOSviewer constructed a co-occurrence map in which different clusters of keywords were observed according to their similarity and co-occurrences. If different terms are used in the same article, then these terms are called co-occurred. In addition, a few keywords are manually added to the final map, because after zooming in the map to use as a figure for this article, some words were vanished from the map due to space limitation. This is the procedure to construct the scientific landscape of HRES-H_2 publications. The magnitude and color of each cluster show the repetition of the keywords and their specified keyword group. The more the distance between two words, the less the two words co-occurred.

Fig. 5 shows the scientific landscape of the publications found in the literature on the topic of HRES-H_2. Three different types of clusters can be observed. The red cluster shows the overall HRES system, economics analysis related to HRES, and global warming issues mitigated by HRES-H_2. Blue cluster represents energy storage devices and algorithms adopted by investigators to tackle problems associated with HRES-H_2. The green cluster indicates fuel cells, software programs used for the simulation of the system, and some advanced optimization techniques. Table 9 outlines the cluster's details.

It is observed from Fig. 5 that all three clusters are interlinked with each other showing that the researchers considered all three areas in their studies, which is obvious. Also, the term optimization lies exactly in the center of these three clusters, which indicates that optimization has been applied with respect to all three cluster terms. In addition, hybrid power system and fuel cells remained in the interest of researchers. The red cluster possesses the keywords which incorporates the whole HRES-H_2 structure along with electricity generation and hydrogen production considering reliability, environmental and economic aspects of the system. On the other hand, blue and green clusters contain the terms associated with the energy conversion and energy storage devices, optimization methods, and software tools. After carefully observing Fig. 5, it can be said that researchers are working on three main topics of HRES-H_2 i.e., (1) techno-economic optimization of HRES-H_2, (2) developing and applying new metaheuristic algorithms for optimization of HRES-H_2, (3) impact

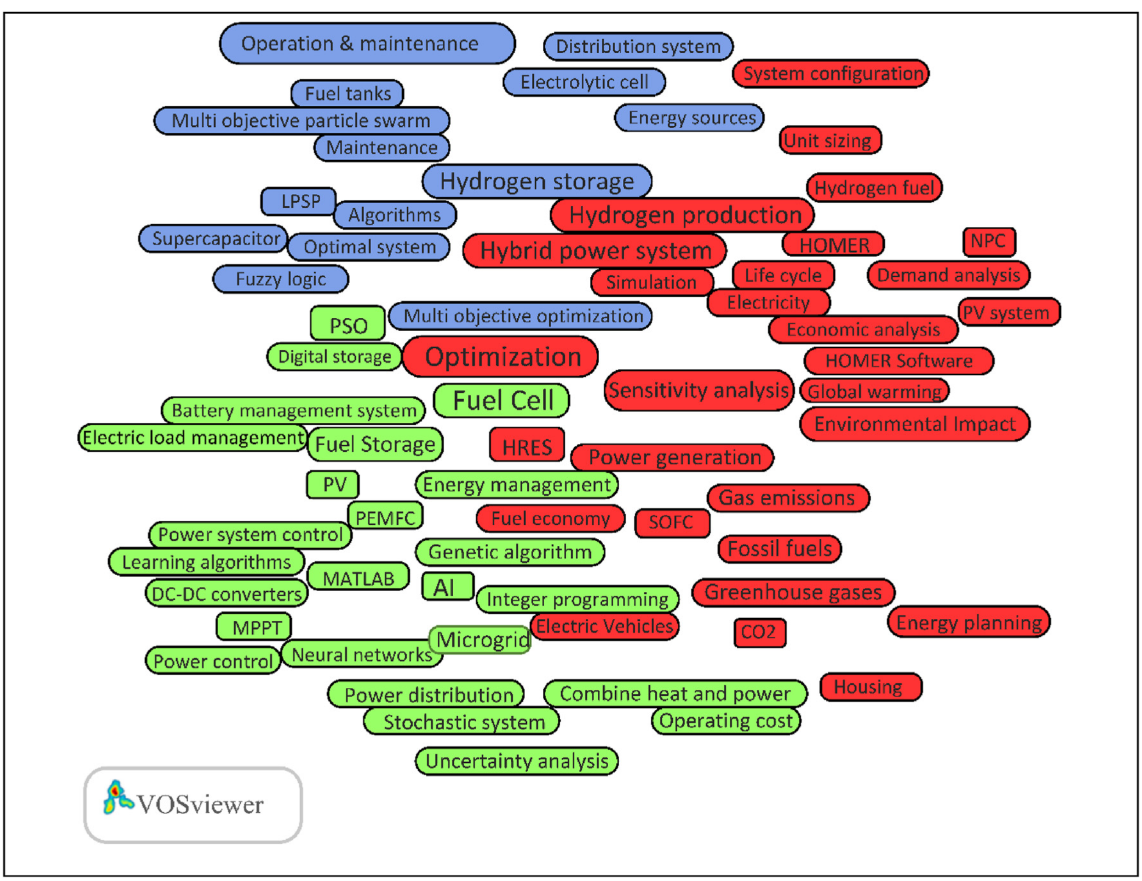

Fig. 5 – Scientific landscape of the publications found in the literature on the topic of HRES-H_2.

Table 9 – Detail of each cluster presented in Fig. 5.

Cluster color	Topics	keywords	No of keywords
Red	Overall HRES system	Hybrid power system, hydrogen production, economic analysis, sensitivity analysis, energy planning, carbon dioxide, optimal system, and simulation.	64
Blue	Energy storage devices	Hydrogen storage, fuel tank, supercapacitor, multi-objective optimization.	49
Green	Fuel cell, optimization techniques	Energy management, learning algorithm, controller, MATLAB, PSO, GA.	28

of renewable and hydrogen technologies on the reduction of global warming. A part from optimization techniques, it is seen that GA and PSO have been used frequently by researchers, while different modern methods for optimization of HRES are found in the existing literature. In addition, keywords related to classical/deterministic and hybrid methods of optimization have not co-occurred frequently in the literature and cannot fulfill the minimum criteria of 10 co-occurrences, therefore, these terms are not visible in the scientific landscape.

Overlay visualization analysis of co-occurred keywords

In order to investigate the recent trends and future scope of HRES-H_2, overlay maps are created based on the timeline of publications and the average number of citations containing co-occurred keywords.

Fig. 6 represents the presence of keywords in the scientific literature view of HRES-H_2 from 2016 to up till now. It can be noticed that from 2019 onwards, researchers put more focus on controlling hydrogen-based hybrid systems and their equipment (terms related to control of a system co-occurred 72 times) which clearly represents that this topic is still under consideration and need breakthrough achievement to discount the total system expenditure and to increase the efficiency. Further, it is observed that sensitivity analysis is being done by researchers in recent years to minimize the risk of energy planning strategies (terms sensitivity analysis, energy planning, and operating cost co-occurred 29, 21, and 20 times respectively). Finally, another research area i.e., hydrogen fuel is found in the scientific landscape which means that the research community of renewable energy is now starting to make efforts to advance the hydrogen technologies for production, storage, and transportation of hydrogen at an economical price (terms hydrogen fuel and fuel economy co-occurred 14 and 11 times). Less number of

Fig. 6 – Presence of keywords in scientific landscape of HRES-H_2 from 2016 to 2020.

co-occurrences of keywords i.e., hydrogen fuel and fuel economy revealed that this topic is not fully explored, therefore, an extensive amount of work needs to be done.

It can also be observed from Fig. 6 that utilizing renewable energy to make the environment polluted free is now also the main area of concern of researchers (terms greenhouse gasses, gas emission, and carbon dioxide co-occurred 16, 10, and 19 times respectively). Less number of co-occurrences shows that there is wide scope for researchers in linking the hybrid renewable energy with hydrogen technologies to mitigate the global warming to a great extent by producing clean electricity and hydrogen which will further fulfill the need of industries, transportation, buildings, etc. So, a combined heat and power system based on renewable energy is the utmost need of the current era. Term CHP co-occurred 11 times, which is clearly inviting researchers to conduct more sophisticated work in this field.

Fig. 7 represents the scientific landscape of keywords that were cited in previous years. It is examined that energy demand analysis is one of the most cited keywords (avg. citations 77.91). Energy demand analysis is necessary when sizing the PV system and wind turbines, which ultimately implies that optimal sizing of HRES (avg. citations 63.50) has been studied widely in the literature. For sizing methods of H_2 based renewable energy system, multi-objective optimization (avg. citations 55.41) and artificial intelligence (avg. citation 53) based methods were more cited representing the focus of researchers on more than one objective function within the same framework of the optimization and intend to make the optimization process more fast, efficient and less prone to error. Table 10 shows the detail of keywords and their total link strength, co-occurrence, and average citations. The total link strength is defined as the total strength of the co-occurrence links of a given keyword with other keywords [195].

Limitations of bibliometric analysis

The software based automated text mining method is greatly reliant on the trait of data gathering. Although the research articles were exported from the search database by relevant search keywords, it is difficult to make sure that all the required research articles have been collected due to synonyms of

Table 10 – Keywords of scientific landscape with high avg. citations observed from Fig. 7.

keywords	Total link strength	Co-occurrence	Avg. citations
Energy demand analysis	171	11	77.91
Hybrid renewable energy	83	10	63.50
Multi-objective optimization	416	39	55.41
Artificial Intelligence	103	10	53

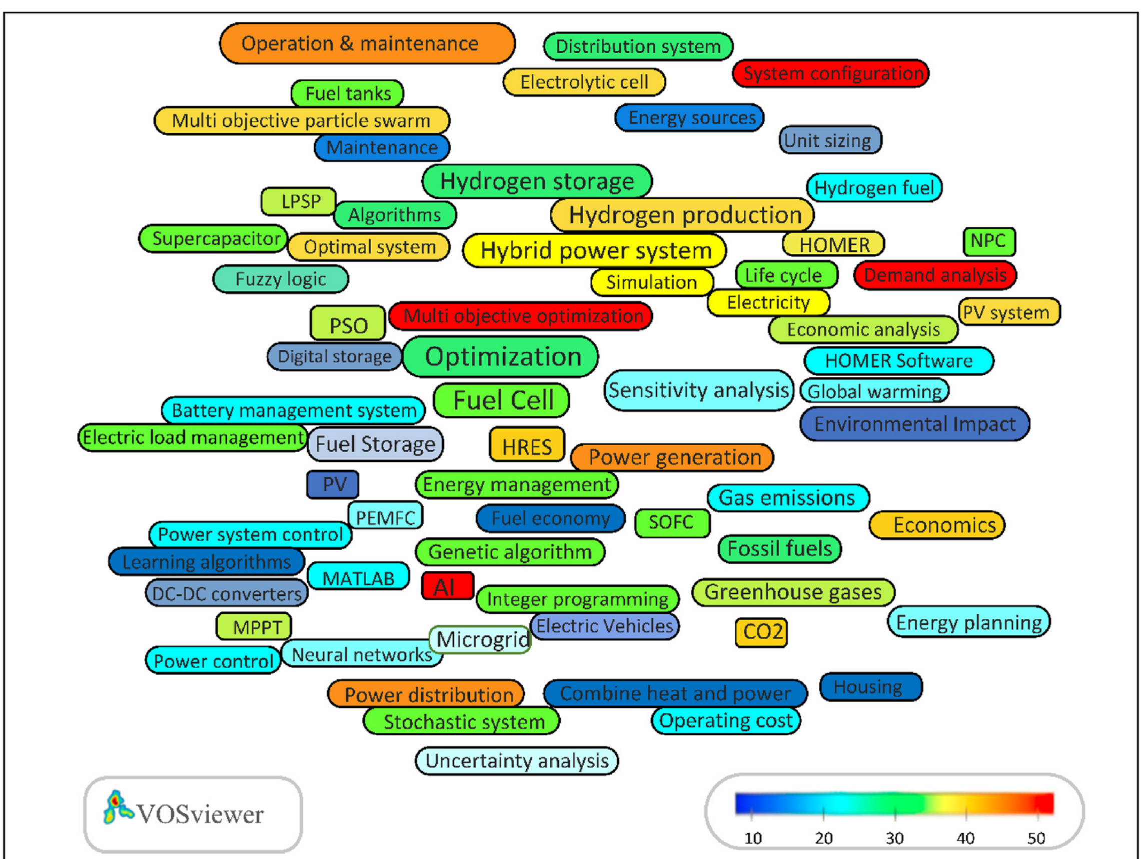

Fig. 7 – Scientific landscape of keywords from corresponding articles that were cited in previous years.

search keywords. The number of keywords would increase unquestionably if research articles in languages other than English are also considered. Further, the bibliometric analysis can be limited in view that the search database engine i.e., Web of Science finds the search terms in articles' title, abstract, and keywords. So, the probability exists that the search keywords lie in the main text of the article. So, it is likely that some of the research articles might be missing. Therefore, a more sophisticated article collection method is required for more accurate bibliometric analysis. Nonetheless, the authors conferred that there is no considerable change in this analysis considering aforesaid limitations.

Critical findings and future recommendations

In recent years, no study is conducted to review the literature on the topic of optimization techniques of HRES with hydrogen technologies. Thus, in this review paper, a first attempt is made in the literature to comprehensively review all the existing optimization approaches adopted by researchers to analyze HRES-H_2. For this purpose, a bibliometric analysis is conducted using a text mining-based software called VOSviewer. To the best of authors knowledge, it is also the first attempt to study HRES-H_2. Various interesting findings have been identified in terms of current trends and future scope, which are explained in the following paragraphs.

To make the theme of the current review more systematic, optimization approaches are broadly categorized into four types i.e., classical approaches, modern approaches, hybrid approaches, and software tools. Classical methods follow a precise methodology to find an optimal solution, however, these approaches have certain limitations such as rigorous iteration process, inflexibility, passive convergence pace, high computational time, not being able to respond to dynamic changes, etc. Contrarily, modern approaches overcome deficiencies of classical methods and possess qualities to find local and global optima with fast convergence speed. These optimization approaches are discussed along with their merits and limitations in Table 2 to have a clear understanding. It is examined that nature inspired meta-heuristic approaches such as PSO, GA, SA, HS, FPA, MBA, ABC, ACO, ICA, etc. (Table 4) are being adopted by researchers across the globe.

A variety of complicated problems cannot be conveniently solved by modern methods to obtain a required optimal solution which opens a new window to introduce a new hybrid optimization algorithm based on two or more modern methods. Therefore, researchers are now focusing to generate hybrid methods having more efficiency, marvelous convergence speed, and more reliable optimal solution. Despite the fact that developing hybrid optimization algorithms offer more complexity as compared to single algorithms, yet few researchers successfully applied hybrid algorithms, for example, SATS, DCHSSA, HCHSA, HFA/HS, and HGWOSCA (Table 6). There is still a need to carry out more efforts to explore and develop sophisticated hybrid optimization methods to compete with the current challenges of hybrid system integration and optimization which shows the future scope of this field.

In literature, to be specifically for the topic of this current review, researchers also used software tools to solve complex optimization problems such as HMOER, RETScreen, TRNSYS, HOGA, etc. The details of these software along with other software tools are mentioned in Table 7 for a better understanding to choose one of them for a particular application and the nature of problem. HOMER is extensively utilized in contrast to other software tools due to its capability to perform optimization of a high number of combinations of HRES.

Clean energy and H_2 production from renewable energy sources and further development in renewable energy technologies show a good potential to meet global energy crises. Literature survey reveals that an enormous number of studies have been done to deploy and optimize HRES-H_2 both in urban and rural areas which shows the increasing trend of utilizing renewable energy to fulfill the global energy demand. However, it is observed that the cost of HRES is not competitive enough in contrast to traditional energy sources. The production of H_2 and its cost is supposed to be one of the potential future research topics. A generous amount of study is needed to increase H_2 production and to make it cost effective, more efficient, and less pollutant so that it can further meet the global energy demand and reduce global warming. A major invention or discovery in this field can transform the way of human life.

For providing electricity to remote and rural areas, mostly standalone HRE systems are employed which often face load profile data issues that ultimately influence the optimization calculation. Thus, more work is needed in this area to accurately forecast and estimate the load profile which is the basis of perfectly sizing the HRES. Accurate modellings of PV, WT, EL, FC, and HT are crucial for optimizing the whole HRES-H_2. The developed models ought to accurately define the energy flow, considering all the parameters including meteorological data involve in power generation and hydrogen production rate.

Apart from the economy of HRES-H_2, the manufacturing and fabrication cost of HRES-H_2 units plays a major role in increasing the overall system price. In solving optimization problems, a smaller number of researchers included social assessment factors in their studies, for example, HDI, employment, and social approval. These factors are generally influenced by the overall cost of the system. It is requisite to considerably reduce the initial system cost as a consequence payback time will be reduced and the rate of investment return would increase, which in turn boost the human development index (HDI) as well as escalate social acceptance.

From the bibliometric analysis of the scientific landscape, the following main research areas are observed:

- Various scientists had considered the cost and reliability objective functions in their optimization studies. Now researchers have started to focus on environmental aspects such as reducing CO_2 by efficiently utilizing HRES with the help of artificial intelligence (AI) based optimization techniques. At present, the hydrogen energy sector/industry is at the strategic turning point of shifting hydrogen from industrial raw materials to large-scale energy development and utilization. There is a huge development space in the future and the relevant industrial chain for hydrogen energy will be greatly developed. Therefore, in this regard, researchers should find ways to advance AI technologies such as big data explosion, improvement in deep learning and machine learning models to make the hydrogen fuel more economical which in turn reduce the GHG. The integration and optimization of HRES-H_2 using AI technologies with the power grid can increase resilience, reliability, and stability of the power system, reduce global warming risk, and mitigate the financial burden on consumers.

- Combined heat and power (CHP) system based on HRES-H_2 has a great potential to fulfill the heat and power demand of multiple sectors i.e., industries, buildings, and transportation. Integration of CHP system with FCEVs is interesting research topic for researchers, keeping in view the user's convenience in supplying heat/cold air. However, the CHP system is not limited to stationary power generation and mobile applications, for instance, vehicles. Power and heat trading with grid and centralized heating system could be another research topic in the energy management strategy for CHP systems based on HRES-$H_{2.}$ Therefore, power trading management in RE-based CHP systems is still attractive for future research and development. Energy management of CHP should be carried out with optimization algorithms to obtain one or more objectives. AI based optimization algorithms can significantly favor CHP system based on HRES-$H_{2.}$
- Technological advancement in fuel cell based electric vehicles is another trending topic among researchers. It is expected to drastically increase the number of FCEVs in the coming years as the technology will continue to mature, causing a revolution, and will eventually be a viable alternative to traditional vehicles. Consequently, the demand for hydrogen will increase, which calls for the studies to produce hydrogen from renewable energy sources. Moreover, keeping in view the storage of the produced hydrogen is a critical issue, therefore, storing hydrogen safely in FCEVs is another research direction. Further research topics in the field of FCEVs could be but are not limited to, cost reduction of FCEVs, advanced multilevel inverters to achieve maximum efficiency, integration of FCEVs with grid, FC waste heat recovery, achieving maximum power from FC using sophisticated optimization algorithms, etc.

The bibliometric analysis clearly shows the future scope of the discussed research fields. New researchers and scientists are recommended to consider these potential gaps to further flourish in these areas of research.

Conclusion

The current review article comprehensively provides an up-to-date literature survey of different types of optimization techniques including classical techniques, modern techniques, hybrid meta-heuristic techniques, and software tools used in HRES-H_2 research and development, which is missing in a recent literature. Each optimization technique has its own distinct convergence pace, veracity level, competency, and in terms of software tools, simulation time, and accuracy. Therefore, choosing an appropriate optimization method is subjected to the type of HRES (stand-alone, grid connected), type of renewable energy sources (PV, WT, Biomass, Hydro, etc.) geographical location, and load demand as well as the type of analysis (economic, reliability, environmental and social). No single optimization approach can perform well enough to deal with all kinds of optimization problems in comparison with other optimization approaches. Nonetheless, modern optimization approaches like meta-heuristic algorithms are found to be more effective as compared to classical algorithms due to the capability of finding local and global optimal and satisfactory calculation efficiency and convergence agility. Moreover, modern hybrid optimization approaches are found to be superior in comparison to modern single optimization approaches, while offering great complexity to develop hybrid optimization code. Although a few studies have adopted hybrid metaheuristic optimization techniques, there is a need to further explore this area and to make the hybrid optimization techniques easy to code and implement while maintaining the same properties. From the software program perspective, HOMER is observed to be widely adopted by researchers due to its good optimization performance, sensitivity analysis, and capability of tackling various complex renewable energy system problems. Finally, the bibliometric analysis, which is a first attempt in the literature to survey the current topic, revealed the current trends and future scope of this field including:

- Development of economical, reliable and sophisticated hydrogen infrastructure to reduce the overall cost of hydrogen fuel including production, storage and transport.
- Mitigate global warming threat by efficiently utilizing HRES-H_2 using AI-based multi-objective optimization techniques such as ANN, deep learning and machine learning models.
- CHP system based on HRES-H_2 to fulfill the power as well as heat demand of industries, buildings, transportation and other sectors of human life. In this context, research focus could be; integrating the CHP system with FCEVs, energy trading of CHP system, energy management of CHP using multi objective optimization algorithms, etc.
- Technological advancement in FCEV to make it more economical, reliable and safe considering research directions such as integrating FCEVs with grid, safety aspects of storing H_2 in FCEVs, maximum power utilization from FC using AI based optimization algorithms, etc.

It can be conclusively said that there is a huge potential for researchers to further explore suggested research areas in detail.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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